AN IMPROVEMENT OF THE FRANÇA–LECLAIR APPROXIMATION OF THE RIEMANN ZEROS WITH A VARIABLE NUMBER OF PRIMES

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Abstract. Let $\gamma_1 \le \gamma_2 \le \cdots$ be the positive ordinates of the nontrivial zeros of $\zeta(s)$, counted with multiplicity, let $N_0(t)$ be the smooth part of the Riemann–von Mangoldt formula and let τ_k be the França–LeClair Lambert clock, $N_0(\tau_k) = k - \frac{1}{2}$. França and LeClair proposed the equation $N_0(t)+\frac{1}{\pi}\arg\zeta(\frac{1}{2}+it) = k-\frac{1}{2}$ for the k-th zero, and LeClair replaced $\arg\zeta$ by the argument of an Euler product over finitely many primes. For a prime cutoff X let $S_X(t) = -\frac{1}{\pi}\sum_{p\le X}\arg(1-p^{-1/2-it})$ and let $u_k(X) = k-\frac{1}{2}-N_0(\gamma_k)-S_X(\gamma_k)$ be the defect with which the true ordinate γ_k satisfies the truncated equation. We prove an exact identity which expresses $\sum_k u_k^2$ through the weighted mean square of $N-N_0-S_X$ minus one twelfth per zero, valid for every C^1 counting model. Combined with Selberg's unconditional approximation of $S(t)$ by prime sums and with the Montgomery–Vaughan mean value theorem it gives, unconditionally and uniformly for $2 \le X \le T^{1/2}$, that the mean of $u_k(X)^2$ over $T < \gamma_k \le 2T$ equals $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O\big(1+(\log\frac{\log T}{\log X})^{1/2}\big)$. Hence any fixed set of primes leaves a defect of mean square $\sim (2\pi^2)^{-1}\log\log T$, and almost all zeros violate the truncated equation by more than any fixed amount, while a cutoff $X = T^c$ that grows with the ordinal keeps the mean-square defect bounded; within $X \le T^{1/2}$ boundedness holds if and only if $\log X \asymp \log T$. We also prove moment formulae for the clock displacement, the continuous analogue of the Selberg–Korolev formulae for Gram intervals, we show that no fixed set of primes changes $\limsup_k|\tau_k-\gamma_k|$, which is at most 0.6331 by the Bellotti–Wong bound, and under a Landau-type hypothesis implied by the Riemann hypothesis we identify the Fourier coefficients of the displacement with the coefficients of the Euler phase. Experiments on the first 2 001 052 zeros show the one-step corrected clock with the ordinal-dependent cutoff $X_k = \max(7,(\tau_k/2\pi)^{0.57})$ reducing the root mean square of $\tau_k-\gamma_k$ in the top window from 0.1789 to 0.0643; the exponent 0.57 is an empirical observation.

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1. Introduction

1.1. **The object.** Let $N(t)$ be the number of zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \beta < 1$ and $0 < \gamma \le t$, counted with multiplicity, and let $\gamma_1 \le \gamma_2 \le \cdots$ be their ordinates listed with multiplicity. The Riemann–von Mangoldt formula writes $N(t)$ as a smooth part plus $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$ [19, §§9.3–9.4]. Inverting the smooth part with the Lambert W function, França and LeClair [4, 5] obtained the explicit sequence

$$\tau_k = \frac{2\pi\,(k - \frac{11}{8})}{W\big((k - \frac{11}{8})/e\big)} \qquad (k \ge 1), \tag{1.1}$$

which we call the clock, and proposed the equation

$$\frac{t}{2\pi}\log\frac{t}{2\pi e} + \frac{1}{\pi}\lim_{\varepsilon\to 0^+}\arg\zeta\big(\tfrac{1}{2} + \varepsilon + it\big) = k - \frac{11}{8} \tag{1.2}$$

for the k-th zero. LeClair [13] replaced $\arg\zeta$ in (1.2) by the argument of an Euler product over N primes, motivated by a random-walk model for the prime sums; he requires $N = N(t) \to \infty$ with $N < [t^2]$ [13, Sect. III], and computes zeros near the ordinals 10^{21}, 10^{22} and 10^{100} with a fixed number $N = 5 \times 10^6$ of primes [13, Sect. VI]. Beyond the growth condition and the upper limit, no rule selecting N from the ordinal is given there, and the validity of the truncated equation is argued from a probabilistic model rather than proved.

The quantity studied here is the defect with which the true ordinate satisfies the truncated equation. For a cutoff $X \ge 2$ let $S_X(t) = -\frac{1}{\pi}\sum_{p \le X}\arg(1 - p^{-1/2-it})$ (full local factors) and

$$u_k(X) = k - \frac{1}{2} - N_0(\gamma_k) - S_X(\gamma_k), \tag{1.3}$$

where N_0 is the smooth count of (2.2). The equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ is (1.2) with $\arg\zeta$ replaced by the truncated Euler product (Lemma 2.6); $u_k(X)$ is measured in units of zeros, so $|u_k(X)| < \frac{1}{2}$ means that the refined count $N_0 + S_X$ assigns to γ_k its correct index.

1.2. **Why it is interesting.** The defect $u_k(X)$ separates two questions which are mixed in numerical comparisons: how well an equation built from finitely many primes describes the zeros, and how well a particular numerical scheme solves that equation. The first question is arithmetic and can be answered with proofs. The answer depends on how the number of primes is chosen, and this is the point at which a rule depending on the ordinal improves on a fixed choice.

1.3. **Main results.** Write $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$. All statements below are proved in the paper; the status of each is indicated.

(i) (Unconditional; Theorem 4.1.) For every C^1 function F and $u_k = k - \frac12 - F(\gamma_k)$, the sum $\sum u_k^2$ over consecutive ordinates equals $\int (N-F)^2\,dF$ minus $\frac{1}{12}$ per zero plus two boundary terms. The same holds for every even power.

(ii) (Unconditional; Theorem 5.1.) The displacement $s_k = k - \frac12 - N_0(\gamma_k)$ has the moments $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} s_k^{2j} = c_j(\log\log T)^j + O_j((\log\log T)^{j-1/2})$ of a centred normal variable of variance $(2\pi^2)^{-1}\log\log T$.

(iii) (Unconditional; Theorem 6.1.) Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\frac{1}{c} + O\Big(1+\sqrt{\log(1/c)}\Big). \tag{1.4}$$

For fixed X this is $\frac{1}{2\pi^2}\log\log T + O(\sqrt{\log\log T})$; for $X = T^c$ with fixed $c \in (0, \frac12]$ it is bounded (Corollaries 6.2–6.4).

(iv) (Unconditional; Proposition 5.4.) $\limsup_k s_k = +\infty$, $\liminf_k s_k = -\infty$, $\limsup_k |\tau_k - \gamma_k| \le 2\pi \cdot 0.10076 < 0.6331$, and a fixed set of primes leaves $\limsup_k |\tau_k^X - \gamma_k|$ unchanged (Proposition 5.5).

(v) (Conditional on Hypothesis 7.1, implied by the Riemann hypothesis; Theorem 7.5.) The mean of $s_k \sin(\gamma_k \log n)$ over $\mathcal{I}_T$ tends to $-\Lambda(n)/(2\pi\sqrt{n}\log n)$, and a finite Euler phase reduces the mean square of s_k by exactly half the sum of its squared coefficients, the coefficient 1 being optimal.

The numerical part (Section 9) uses the one-step corrected clock $\tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$ with an ordinal-dependent cutoff on the first 2 001 052 zeros. The exponent 0.57 of the cutoff used there is an empirical observation; a linearized model for it is stated as a heuristic, and its quantitative failure is recorded (Remark 8.3).

1.4. **Relation to Selberg's work.** Selberg's approximation of $S(t)$ by sums over primes up to a power of T is a statement about $S(t)$ averaged over t [17, 18]; we use it in the form proved in [6, §3]. It does not concern the ordinates γ_k or the clock. The passage from t-averages to averages over the zeros is made here by the exact identity of Theorem 4.1, which has no error term and does not require bounds for the number of zeros in short intervals.

2. Notation and the clock

2.1. **Counting functions.** Let $\vartheta(t) = \Im\log\Gamma(\frac14 + \frac{it}{2}) - \frac{t}{2}\log\pi$ be the Riemann–Siegel theta function. For $t > 0$ not an ordinate [19, Thms. 9.3–9.4],

$$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t), \qquad S(t) = \frac{1}{\pi}\arg\zeta\big(\tfrac12 + it\big) = O(\log t), \tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 along the segment to $2+it$ and then horizontally to $\frac12 + it$. We set

$$\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}, \qquad N_0(t) = \frac{1}{\pi}\Theta_0(t) + 1, \qquad \omega(t) = \vartheta(t) - \Theta_0(t), \tag{2.2}$$

so that $\omega(t) = \frac{1}{48t} + O(t^{-3})$ by Stirling's formula and

$$N(t) = N_0(t) + S(t) + \frac{\omega(t)}{\pi}. \tag{2.3}$$

The derivative of the smooth count is

$$\mathcal{L}(t) = N_0'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{2.4}$$

For $T \geq 1$ let $\mathcal{I}_T = \{k \geq 1 : T < \gamma_k \leq 2T\}$. Since $N(2T) - N(T) = N_0(2T) - N_0(T) + O(\log T)$ by (2.1),

$$\#\mathcal{I}_T = T\mathcal{L}(T) + O(T). \tag{2.5}$$

2.2. **The clock and the displacements.**

Lemma 2.1. *The sequence* (1.1) *is the unique solution of* $N_0(\tau_k) = k - \frac{1}{2}$ *with* $\tau_k > 2\pi$, $k \geq 2$. *As a function of a real variable* k, τ_k *is smooth and increasing with*

$$\frac{d\tau_k}{dk} = \frac{1}{\mathcal{L}(\tau_k)} = \frac{2\pi}{\log(\tau_k/2\pi)}. \tag{2.6}$$

Proof. With $x = t/2\pi$ the equation $N_0(t) = k - \frac{1}{2}$ reads $x\log(x/e) = k - \frac{11}{8}$. Putting $x = ey$ gives $y\log y = (k - \frac{11}{8})/e$, so $\log y = W((k - \frac{11}{8})/e)$ with the principal branch [3] and $x = (k - \frac{11}{8})/W((k - \frac{11}{8})/e)$. The map $x \mapsto x\log(x/e)$ is increasing for $x > 1$, which gives uniqueness, and (2.6) follows by differentiating $N_0(\tau_k) = k - \frac{1}{2}$. □

Definition 2.2. The ordinal displacement and the displacement of the clock are

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad \delta_k = \tau_k - \gamma_k. \tag{2.7}$$

By Lemma 2.1, $s_k = N_0(\tau_k) - N_0(\gamma_k)$, and the mean value theorem gives

$$s_k = \mathcal{L}(\xi_k)\,\delta_k \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{2.8}$$

Remark 2.3. Since $\Theta_0(\tau_k) = \pi(k - \frac{3}{2})$ and $\vartheta = \Theta_0 + \omega$, the clock point τ_k is the point at which ϑ takes the value $\pi(k - \frac{3}{2})$, up to a shift $O(\tau_k^{-1}(\log\tau_k)^{-1})$. Thus τ_k lies, in phase, half way between the Gram points g_{k-2} and g_{k-1}, defined by $\vartheta(g_n) = n\pi$. The integer displacement of γ_k with respect to the Gram intervals is the object of Gram's law; see Remark 5.3.

Lemma 2.4. *For every* k *with* $\gamma_k > 2$,

$$N(\gamma_k^-) - N_0(\gamma_k) + \frac{1}{2} \leq s_k \leq N(\gamma_k^+) - N_0(\gamma_k) - \frac{1}{2}, \tag{2.9}$$

where $N(\gamma^\pm)$ *are the one-sided limits. In particular* $|s_k| \leq |S(\gamma_k^-)| + |S(\gamma_k^+)| + 1$.

Proof. The indices j with $\gamma_j = \gamma_k$ are $N(\gamma_k^-) + 1, \ldots, N(\gamma_k^+)$, hence $N(\gamma_k^-) + 1 \leq k \leq N(\gamma_k^+)$; subtract $\frac{1}{2} + N_0(\gamma_k)$ and use (2.3) with $|\omega|/\pi < \frac{1}{2}$. □

2.3. **Euler phases and the truncated equation.** For $n \geq 2$ put

$$a_n = -\frac{\Lambda(n)}{\pi\sqrt{n}\,\log n}, \tag{2.10}$$

where Λ is the von Mangoldt function. For a set A of prime powers with $\sum_{n\in A}|a_n| < \infty$ let

$$S_A(t) = \sum_{n\in A} a_n \sin(t\log n), \qquad v_A = \frac{1}{2}\sum_{n\in A} a_n^2. \tag{2.11}$$

For $X \geq 2$ let $A_X = \{p^m : p \leq X,\ m \geq 1\}$ and $S_X = S_{A_X}$. Since $\arg(1-z) = -\Im\sum_{m\geq 1} z^m/m$ for $|z| < 1$,

$$S_X(t) = -\frac{1}{\pi}\sum_{p\leq X}\arg(1 - p^{-1/2-it}) = -\frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{\sin(mt\log p)}{m\,p^{m/2}}, \tag{2.12}$$

with the principal argument. For a finite set of primes P we write S_P for the same sum over $p \in P$, and

$$c_P = \frac{1}{2\pi^2} \sum_{p \in P} \mathrm{Li}_2\Big(\frac{1}{p}\Big) = \lim_{M \to \infty} v_{\{p^m : p \in P,\ m \le M\}}, \tag{2.13}$$

where $\mathrm{Li}_2(x) = \sum_{m \ge 1} x^m/m^2$. We also use the bounds

$$|S_X(t)| \le \sigma_X := \frac{1}{\pi} \sum_{p \le X} \sum_{m \ge 1} \frac{1}{m\, p^{m/2}}, \qquad |S_X'(t)| \le M_X := \frac{1}{\pi} \sum_{p \le X} \frac{\log p}{\sqrt{p} - 1}, \tag{2.14}$$

and note that $\sigma_X \ll \sqrt{X}/\log X$ and $M_X \ll \sqrt{X}$ by the prime number theorem.

Definition 2.5. For $X \ge 2$ the truncated counting model and the defect of the k-th zero are

$$F_X(t) = N_0(t) + S_X(t), \qquad u_k(X) = k - \frac{1}{2} - F_X(\gamma_k) = s_k - S_X(\gamma_k). \tag{2.15}$$

The one-step corrected clock is

$$\tau_k^X = \tau_k - \frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)} = \tau_k - \frac{d\tau_k}{dk}\, S_X(\tau_k), \tag{2.16}$$

and, for a saturation level $B > 0$, $\tau_k^{X,B} = \tau_k - \mathrm{sat}_B(S_X(\tau_k)/\mathcal{L}(\tau_k))$ with $\mathrm{sat}_B(y) = \mathrm{sgn}(y)\min(|y|, B)$. We write $\delta_k^X = \tau_k^X - \gamma_k$ and $\delta_k^{X,B} = \tau_k^{X,B} - \gamma_k$.

Lemma 2.6. *Equation* (1.2) *is equivalent to* $N_0(t) + S(t) = k - \frac{1}{2}$. *Replacing* $\frac{1}{\pi}\arg\zeta(\frac{1}{2} + it)$ *by* $\frac{1}{\pi}\arg\prod_{p \le X}(1 - p^{-1/2-it})^{-1}$ *turns it into* $F_X(t) = k - \frac{1}{2}$, *and* τ_k^X *is the first step of the iteration* $t \mapsto t - (F_X(t) - k + \frac{1}{2})/\mathcal{L}(t)$ *started at* $t = \tau_k$.

Proof. By (2.2), $\frac{t}{2\pi}\log\frac{t}{2\pi e} = N_0(t) - \frac{7}{8}$, so (1.2) is $N_0(t) + S(t) = k - \frac{1}{2}$. The argument of $\prod_{p \le X}(1 - p^{-1/2-it})^{-1}$ is $-\sum_{p \le X}\arg(1 - p^{-1/2-it}) = \pi S_X(t)$ by (2.12). The last statement follows from $F_X(\tau_k) - k + \frac{1}{2} = S_X(\tau_k)$. □

3. Known results used

This section collects standard material; nothing in it is new.

Theorem 3.1 (Riemann–von Mangoldt; Littlewood; [19, Thms. 9.4, 9.9 and §9.4], [14])**.** $S(t) = O(\log t)$*; the gaps* $\gamma_{k+1} - \gamma_k$ *are bounded; and* $S_1(t) = \int_0^t S(u)\,du = O(\log t)$.

Theorem 3.2 (Bellotti–Wong [1])**.** *For* $T \ge e$,

$$\Big|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\Big| \le 0.10076\log T + 0.24460\log\log T + 8.08344. \tag{3.1}$$

Earlier explicit forms are due to Trudgian [20] and Hasanalizade, Shen and Wong [10].

Theorem 3.3 (Selberg [17]; see [6, p. 225] and [19, §9.25])**.** *For every integer* $j \ge 1$, *with* $c_j = \frac{(2j)!}{j!\,(2\pi)^{2j}}$,

$$\int_T^{2T} S(t)^{2j}\,dt = c_j\, T(\log\log T)^j + O_j\big(T(\log\log T)^{j-1/2}\big). \tag{3.2}$$

In [6, p. 225] the formula is quoted for $\int_T^{T+H}$ with $T^{1/2+a} \le H \le T$; we use $H = T$.

Theorem 3.4 (Selberg's approximation, in the form of [6, §3, pp. 232–233]; [18])**.** *Let* $x_0 = T^{1/160}$ *and* $S_{x_0}^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p < x_0} p^{-1/2}\sin(t\log p)$. *Then*

$$\int_T^{2T} \big(S(t) - S_{x_0}^{\mathrm{pr}}(t)\big)^4 dt \ll T. \tag{3.3}$$

Reference to the proof. In [6, §3], for a primitive character χ modulo q and $x \geq 2$, Selberg's expression $S(t,\chi) - \frac{1}{\pi}\Im \sum_{p<x^3} \chi(p)p^{-1/2-it} = \sum_{i=1}^{5} O(B_i(t))$ is written with explicit functions $B_1, \dots, B_5$, and it is shown on p. 233 that $\|B_i\| \ll k^2 H^{1/2k}$ for $x = T^{a_2/60k}$, where $\|f\| = (\int_T^{T+H} |f|^{2k})^{1/2k}$ and the hypotheses are those of the Main Theorem of [6]: $0 < a_1, a_2 \leq \frac{1}{4}$, $q \leq T^{1/4-a_1}$ and $T^{1/2+a_2} \leq H \leq T$. Take χ principal ($q = 1$, $\zeta = L(s,\chi_0)$), $k = 2$, $a_2 = \frac{1}{4}$ and $H = T$. Then $x^3 = T^{1/160}$, and $\frac{1}{\pi}\Im p^{-1/2-it} = -\frac{1}{\pi} p^{-1/2}\sin(t\log p)$. □

Theorem 3.5 (Montgomery–Vaughan [16, Cor. 3])**.** *For complex b_n, $\int_T^{2T} \left|\sum_{n\leq Y} b_n n^{-it}\right|^2 dt = \sum_{n\leq Y} |b_n|^2 (T + O(n))$ with an absolute implied constant.*

The version on $[T, 2T]$ follows from the one on $[0, T]$ by replacing b_n by $b_n n^{-iT}$.

Theorem 3.6 (Mertens [15])**.** *For $y \geq 2$, $\sum_{p\leq y} \frac{\log p}{p} = \log y + O(1)$ and $\sum_{p\leq y} \frac{1}{p} = \log\log y + B_1 + O(\frac{1}{\log y})$ with a constant B_1.*

Theorem 3.7 (Kusmin–Landau; [9, Thm. 2.1])**.** *Let f be real with a continuous monotone derivative on $[a, b]$, and suppose that $\|f'(x)\| \geq \lambda > 0$ there, where $\|\cdot\|$ is the distance to the nearest integer. Then $\sum_{a<n\leq b} e^{2\pi i f(n)} \ll \lambda^{-1}$.*

Theorem 3.8 (Landau [12]; uniform version by Gonek [7])**.** *For fixed $x > 1$, $\sum_{0<\gamma\leq T} x^\rho = -\frac{T}{2\pi}\Lambda(x) + O(\log T)$, where $\Lambda(x) = 0$ unless x is a prime power, and the zeros are counted with multiplicity.*

Theorem 3.9 (Littlewood; [19, Thm. 14.13])**.** *On the Riemann hypothesis, $S(t) = O(\log t/\log\log t)$.*

4. Exact identities at the zeros

The results of this section are original. They are exact and involve no hypothesis on the zeros, not even their simplicity.

Theorem 4.1. *Let $m < n$ be indices, let F be a real C^1 function on $[\gamma_m, \gamma_n]$ and put $u_k = k - \frac{1}{2} - F(\gamma_k)$ for $m \leq k \leq n$. For every integer $j \geq 1$,*

$$\int_{\gamma_m}^{\gamma_n} \big(N(t) - F(t)\big)^{2j} F'(t)\,dt = \sum_{k=m+1}^{n-1} P_j(u_k) + \frac{(\frac{1}{2}+u_m)^{2j+1} + (\frac{1}{2}-u_n)^{2j+1}}{2j+1}, \tag{4.1}$$

where

$$P_j(u) = \frac{(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}}{2j+1} = \sum_{i=0}^{j} \binom{2j+1}{2i} \frac{2^{2i-2j}}{2j+1}\, u^{2i} = u^{2j} + \sum_{i<j} b_{j,i} u^{2i}. \tag{4.2}$$

In particular, for $j = 1$,

$$\sum_{k=m+1}^{n-1} u_k^2 = \int_{\gamma_m}^{\gamma_n} \big(N(t) - F(t)\big)^2 F'(t)\,dt - \frac{n-m-1}{12} - \frac{(\frac{1}{2}+u_m)^3 + (\frac{1}{2}-u_n)^3}{3}. \tag{4.3}$$

Proof. Fix k with $m \leq k < n$. If $\gamma_k < \gamma_{k+1}$ then exactly k ordinates, counted with multiplicity, are $\leq t$ for $t \in (\gamma_k, \gamma_{k+1})$, so $N(t) = k$ there. The substitution $y = F(t)$ is valid for every C^1 function, monotone or not, and gives

$$\int_{\gamma_k}^{\gamma_{k+1}} \big(k - F(t)\big)^{2j} F'(t)\,dt = \int_{F(\gamma_k)}^{F(\gamma_{k+1})} (k-y)^{2j}\,dy = \frac{(\frac{1}{2}-u_{k+1})^{2j+1} + (\frac{1}{2}+u_k)^{2j+1}}{2j+1}, \tag{4.4}$$

because $F(\gamma_k) - k = -\frac{1}{2} - u_k$ and $F(\gamma_{k+1}) - k = \frac{1}{2} - u_{k+1}$. If $\gamma_k = \gamma_{k+1}$ the left side of (4.4) is 0; since then $u_{k+1} = u_k + 1$, also the right side is $(-\frac{1}{2}-u_k)^{2j+1} + (\frac{1}{2}+u_k)^{2j+1} = 0$. The values of N at the finitely many ordinates do not affect the integrals. Summing (4.4) over $m \leq k < n$, the

term $(\frac12+u_k)^{2j+1}$ with $m<k<n$ meets the term $(\frac12-u_k)^{2j+1}$ coming from the gap (γ_{k-1},γ_k), which gives (4.1). The expansion (4.2) keeps the even powers of the binomial expansion, and the coefficient of u^{2j} is $\binom{2j+1}{2j}/(2j+1)=1$. For $j=1$, $P_1(u)=u^2+\frac{1}{12}$. □

Remark 4.2. The constant $\frac{1}{12}$ is the mean square of a centred sawtooth of unit step: between two consecutive zeros $N-F$ decreases through one unit in the variable $y=F(t)$. Identity (4.3) therefore says that the mean square of the defect at the zeros is the dF-mean square of $N-F$ minus the contribution of the sawtooth. Nothing is assumed about the sign of F'; when F' changes sign the identity remains exact, with the integral counted with sign.

Corollary 4.3. *With $F=N_0$, so that $u_k=s_k$ and $N-N_0=S+\omega/\pi$ almost everywhere,*

$$\sum_{k=m+1}^{n-1} s_k^2=\int_{\gamma_m}^{\gamma_n}\Big(S(t)+\frac{\omega(t)}{\pi}\Big)^2\mathcal{L}(t)\,dt-\frac{n-m-1}{12}-\frac{(\frac12+s_m)^3+(\frac12-s_n)^3}{3}. \tag{4.5}$$

With $F=F_X$, so that $u_k=u_k(X)$ and $N-F_X=S-S_X+\omega/\pi$ almost everywhere,

$$\sum_{k=m+1}^{n-1} u_k(X)^2=\int_{\gamma_m}^{\gamma_n}\Big(S(t)-S_X(t)+\frac{\omega(t)}{\pi}\Big)^2\big(\mathcal{L}(t)+S_X'(t)\big)dt-\frac{n-m-1}{12}-\frac{(\frac12+u_m)^3+(\frac12-u_n)^3}{3}. \tag{4.6}$$

Proof. Theorem 4.1 with $j=1$ and (2.3), (2.15). □

Lemma 4.4. *Let $2<a<b$ be points which are not ordinates, let $f\in C^1[a,b]$ and $G=N-N_0$. Then*

$$\sum_{a<\gamma_k\le b} s_k f(\gamma_k)=\int_a^b G(t)f(t)\mathcal{L}(t)\,dt+\frac12\Big[f\,G^2\Big]_a^b-\frac12\int_a^b G(t)^2f'(t)\,dt. \tag{4.7}$$

Proof. Let γ be an ordinate of multiplicity ν and $G^*(\gamma)=\frac12(G(\gamma^-)+G(\gamma^+))$. The indices with $\gamma_k=\gamma$ are $N(\gamma^-)+1,\dots,N(\gamma^-)+\nu$, so $\sum_{\gamma_k=\gamma}(k-\frac12)=\nu N(\gamma^-)+\frac{\nu^2}{2}=\nu\,\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and $\sum_{\gamma_k=\gamma}s_k=\nu\,G^*(\gamma)$. Hence the left side of (4.7) is the Stieltjes integral $\int_{(a,b]}G^*f\,dN$. As measures $dN=dG+\mathcal{L}\,dt$. The function G is C^1 between ordinates and jumps by ν at an ordinate of multiplicity ν; at a jump $\Delta(G^2)=2G^*\Delta G$, and between jumps $(G^2)'=2GG'$. Hence $G^*\,dG=\frac12 d(G^2)$, and integration by parts gives $\int_{(a,b]}f\,d(G^2)=[fG^2]_a^b-\int_a^b G^2f'\,dt$. □

Corollary 4.5. $\sum_{\gamma_k\le t}s_k=O(\log^2 t)$. *In particular the mean of s_k over $k\le K$ is $O(K^{-1}\log^2 K)$.*

Proof. Apply Lemma 4.4 with $f\equiv1$, $a=14$ (below γ_1) and $b=t$. Then $G^2=O(\log^2 t)$ by Theorem 3.1, and $\int_{14}^t G\mathcal{L}\,du=\int_{14}^t S\mathcal{L}\,du+O(1)=\mathcal{L}(t)S_1(t)-\int_{14}^t S_1(u)\frac{du}{2\pi u}+O(\log t)=O(\log^2 t)$ by $S_1=O(\log t)$. □

5. Moments and extremes of the clock displacement

Theorem 5.1. *For every fixed integer $j\ge1$, as $T\to\infty$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}s_k^{2j}=c_j(\log\log T)^j+O_j\big((\log\log T)^{j-1/2}\big),\qquad c_j=\frac{(2j)!}{j!\,(2\pi)^{2j}}. \tag{5.1}$$

In particular the mean square of s_k over $\mathcal{I}_T$ is $\frac{1}{2\pi^2}\log\log T+O(\sqrt{\log\log T})$ and

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\delta_k^2=\frac{2\log\log T}{\log^2 T}\big(1+O((\log\log T)^{-1/2})\big). \tag{5.2}$$

Proof. Let m and n be the smallest and largest indices in $\mathcal{I}_T$. Theorem 4.1 with $F = N_0$ gives

$$\sum_{k=m+1}^{n-1} P_j(s_k) = \int_{\gamma_m}^{\gamma_n} \Big(S + \frac{\omega}{\pi}\Big)^{2j} \mathcal{L}\, dt + O(\log^{2j+1} T), \tag{5.3}$$

since $|s_k| \ll \log T$ by Lemma 2.4. The intervals $[T, \gamma_m]$ and $[\gamma_n, 2T]$ have bounded length (Theorem 3.1) and the integrand is $O(\log^{2j+1} T)$ there; moreover $|(S+\omega/\pi)^{2j} - S^{2j}| \ll T^{-1}\log^{2j} T$. Since $\mathcal{L}(t) = \mathcal{L}(T) + O(1)$ on $[T, 2T]$, Theorem 3.3 yields

$$\int_T^{2T} S^{2j}\mathcal{L}\, dt = \mathcal{L}(T)\, c_j T(\log\log T)^j + O(\mathcal{L}(T)\, T(\log\log T)^{j-1/2}). \tag{5.4}$$

Adding the terms $k = m, n$, which are $O(\log^{2j} T)$, and dividing by (2.5),

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} P_j(s_k) = c_j(\log\log T)^j + O((\log\log T)^{j-1/2}). \tag{5.5}$$

For $j = 1$ this is (5.1), because $P_1(s) = s^2 + \frac{1}{12}$. If (5.1) holds for all $i < j$, then $\sum_{i<j} b_{j,i} \frac{1}{\#\mathcal{I}_T} \sum s_k^{2i} = O((\log\log T)^{j-1})$ and (5.1) follows for j. For (5.2) use (2.8): $\mathcal{L}(\xi_k) = \mathcal{L}(T)(1 + O(1/\log T))$ because $|\delta_k| \ll 1$, and $\mathcal{L}(T)^2 = (2\pi)^{-2}\log^2 T\,(1 + O(1/\log T))$. □

Corollary 5.2. *Let $\sigma_T^2 = \frac{1}{2\pi^2}\log\log T$ and let Z be a standard normal variable. For every fixed $\eta \geq 0$,*

$$\frac{1}{\#\mathcal{I}_T} \#\{k \in \mathcal{I}_T : |s_k| \leq \eta\, \sigma_T\} \longrightarrow \mathbb{P}(|Z| \leq \eta). \tag{5.6}$$

In particular, for every fixed $A > 0$, $\#\{k \in \mathcal{I}_T : |s_k| \leq A\} = o(\#\mathcal{I}_T)$.

Proof. By Theorem 5.1 the empirical distributions of s_k^2/σ_T^2, $k \in \mathcal{I}_T$, have j-th moments tending to $c_j(2\pi^2)^j = \frac{(2j)!}{j!\,2^j} = \mathbb{E}[Z^{2j}]$. The law of Z^2 is determined by its moments, its moment generating function being finite near 0, so the method of moments [2, §30] gives convergence in distribution of s_k^2/σ_T^2 to Z^2; the limit law is continuous. □

Remark 5.3. Selberg stated, and Korolev proved [11], moment formulae for the integer displacement of γ_n with respect to the Gram intervals, and Korolev deduced that γ_n lies in its own Gram interval only for a set of indices of density zero. By Remark 2.3, Theorem 5.1 and Corollary 5.2 are the analogous statements for the real displacement s_k with respect to the half-integer Gram points τ_k. The proof given here is a direct consequence of Theorem 4.1 and Selberg's moments.

Proposition 5.4. *(a)* $\limsup_{k\to\infty} s_k = +\infty$ *and* $\liminf_{k\to\infty} s_k = -\infty$.
(b) $\limsup_{k\to\infty} |\delta_k| \leq 2\pi \cdot 0.10076 = 0.633093\ldots$.
(c) On the Riemann hypothesis, $\delta_k \to 0$.

Proof. (a) Let $\gamma_k < \gamma_{k+1}$ and $t \in (\gamma_k, \gamma_{k+1})$. Then $N(t) = k$ and, by monotonicity of N_0,

$$s_{k+1} - \frac{1}{2} \leq N(t) - N_0(t) \leq s_k + \frac{1}{2}. \tag{5.7}$$

Suppose $s_k \leq A$ for all large k. Then $S(t) \leq A + 1$ for all large t by (2.3). By Theorem 3.1, $\int_T^{2T} S\, dt = O(\log T)$, so $\int_T^{2T} S^-\, dt = \int_T^{2T} S^+ dt - \int_T^{2T} S\, dt \leq (A+1)T + O(\log T)$ and $\int_T^{2T} |S|\, dt \ll T$. Hölder's inequality with exponents $\frac{3}{2}$ and 3 and Theorem 3.3 with $j = 2$ give

$$\int_T^{2T} S^2 dt \leq \Big(\int_T^{2T} |S|\, dt\Big)^{2/3} \Big(\int_T^{2T} S^4 dt\Big)^{1/3} \ll T(\log\log T)^{2/3}, \tag{5.8}$$

which contradicts (3.2) with $j = 1$. The statement for lim inf follows in the same way from the left inequality in (5.7).

(b) By Lemma 2.4, $|s_k| \le \max_\pm |N(\gamma_k^\pm) - N_0(\gamma_k)| + \frac{1}{2}$, and $N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}$. Theorem 3.2 gives $|s_k| \le 0.10076\log\gamma_k + O(\log\log\gamma_k)$. By (2.8), $|\delta_k| = |s_k|/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{1}{2\pi}\log\gamma_k\,(1 + o(1))$.

(c) Theorem 3.9 and Lemma 2.4 give $s_k = O(\log\gamma_k/\log\log\gamma_k)$, hence $\delta_k = O(1/\log\log\gamma_k)$. □

Proposition 5.5. *Let $X \ge 2$ be fixed and $B \in (0,\infty]$. Then $\limsup_k |\delta_k^{X,B}| = \limsup_k |\delta_k|$, while $\limsup_k |u_k(X)| = \infty$.*

Proof. $|\delta_k^{X,B} - \delta_k| \le |S_X(\tau_k)|/\mathcal{L}(\tau_k) \le \sigma_X/\mathcal{L}(\tau_k) \to 0$ by (2.14), and $|u_k(X) - s_k| \le \sigma_X$; apply Proposition 5.4(a). □

6. The main theorem: a variable cutoff in the truncated equation

6.1. **Statement.**

Theorem 6.1. *There are absolute constants T_0 and C such that for $T \ge T_0$ and every X with $2 \le X \le T^{1/2}$, writing $c = \log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \le C\Big(1 + \sqrt{\log(1/c)}\Big). \tag{6.1}$$

The proof is given in Subsection 6.3. Note that $\log(1/c) = \log\log T - \log\log X$.

Corollary 6.2 (fixed cutoff)**.** *Let $X \ge 2$ be fixed. Then*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\log T + O_X\big(\sqrt{\log\log T}\big), \tag{6.2}$$

and for every fixed $A > 0$, $\#\{k \in \mathcal{I}_T : |u_k(X)| \le A\} = o(\#\mathcal{I}_T)$.

Proof. The first statement is (6.1). For the second, $|u_k(X) - s_k| \le \sigma_X$, so $|u_k(X)| \le A$ implies $|s_k| \le A + \sigma_X$; apply Corollary 5.2. □

Corollary 6.3 (power cutoff)**.** *Let $0 < c \le \frac{1}{2}$ be fixed and $X = T^c$. Then*

$$\limsup_{T\to\infty} \frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(T^c)^2 \le \frac{1}{2\pi^2}\log\frac{1}{c} + C\Big(1 + \sqrt{\log(1/c)}\Big), \tag{6.3}$$

and for every function $\eta(T) \to \infty$, $\#\{k \in \mathcal{I}_T : |u_k(T^c)| > \eta(T)\} = o(\#\mathcal{I}_T)$. The same holds for $X = K^c$ with $K = N(T)$ and fixed $0 < c < \frac{1}{2}$, because then $X = T^{c_T}$ with $c_T = c + O(\log\log T/\log T)$, so that $c_T < \frac{1}{2}$ for large T and $\log(1/c_T) = \log(1/c) + o(1)$.

Proof. The bound is (6.1); the second statement follows by Chebyshev's inequality. □

Corollary 6.4 (dichotomy)**.** *Let $X = X(T)$ with $2 \le X \le T^{1/2}$. The mean square of $u_k(X)$ over $k \in \mathcal{I}_T$ is bounded as $T \to \infty$ if and only if $\liminf_{T\to\infty} \log X/\log T > 0$. In particular, for any fixed $X_0 \ge 2$ and fixed $c \in (0, \frac{1}{2}]$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(u_k(X_0)^2 - u_k(T^c)^2\Big) = \frac{1}{2\pi^2}\log\log T + O\big(\sqrt{\log\log T}\big) \longrightarrow \infty. \tag{6.4}$$

Proof. Both directions follow from (6.1), since $x \mapsto \frac{x}{2\pi^2} - C(1 + \sqrt{x})$ tends to infinity. For (6.4) subtract (6.3) from (6.2). □

Remark 6.5. Corollary 6.4 is the precise sense in which an ordinal-dependent number of primes improves the truncated França–LeClair equation. With $N = \#\{p \le X\}$ primes, $\log N = \log X + O(\log\log X)$, so the condition is that $\log N$ grows proportionally to $\log T$. A fixed N, as in the computations of [13, Sect. VI], leaves a defect whose mean square grows like $(2\pi^2)^{-1}\log\log T$, and almost all zeros then violate the truncated equation by more than any fixed amount. LeClair's

requirement $N(t)\to\infty$ [13, Sect. III] is necessary but not sufficient: if $\log N = o(\log T)$, the mean-square defect still tends to infinity. The range $c>\frac12$, which contains LeClair's upper limit $N<[t^2]$, is not covered by Theorem 6.1.

Remark 6.6. Theorem 6.1 concerns a cutoff that is constant on each window $\mathcal{I}_T$. Taking $T=2^r$ and $X=2^{rc}$ on $\mathcal{I}_{2^r}$ gives a cutoff depending on the ordinal through the window; the rule $X_k=(\tau_k/2\pi)^\theta$ used in Section 9 differs from such a block rule by a factor at most 2^θ inside each window. The extension of (6.1) to cutoffs that vary from zero to zero is not proved here (Question 11.3).

6.2. **Lemmas on prime polynomials.** Fix T large and $2\le X\le T^{1/2}$, let $x_0=T^{1/160}$ and decompose $S_X=S_X^{(1)}+S_X^{(2)}+S_X^{(\ge3)}$ according to $m=1$, $m=2$ and $m\ge3$ in (2.12). Let $\mathcal{P}$ be the set of primes in (X,x_0) if $X<x_0$, and in $[x_0,X]$ if $X\ge x_0$, let $\epsilon=+1$ in the first case and $\epsilon=-1$ in the second, and put

$$D_1(t)=-\frac{1}{\pi}\sum_{p\in\mathcal{P}}\frac{\sin(t\log p)}{\sqrt{p}},\qquad V=\sum_{p\in\mathcal{P}}\frac{1}{p},\qquad D=S_{x_0}^{\mathrm{pr}}-S_X=\epsilon D_1-S_X^{(2)}-S_X^{(\ge3)}. \tag{6.5}$$

Lemma 6.7. *$V=\log(1/c)+O(1)$ uniformly for $0<c\le\frac12$ with $X=T^c\ge2$.*

Proof. By Theorem 3.6, $V=|\log\log x_0-\log\log X|+O(1)=|\log\frac{1}{160c}|+O(1)$. If $c\le\frac{1}{160}$ this is $\log\frac1c-\log160+O(1)$; if $\frac{1}{160}<c\le\frac12$ both V and $\log\frac1c$ are $O(1)$. □

Lemma 6.8. *For real b_n, $2\le n\le Y$,*

$$\int_T^{2T}\Big(\sum_{n\le Y}b_n\sin(t\log n)\Big)^2dt=\frac{T}{2}\sum_{n\le Y}b_n^2+O\Big(\sum_{n\le Y}nb_n^2+\Big(\sum_{n\le Y}|b_n|\Big)^2\Big), \tag{6.6}$$

and the same holds with cos *in place of* sin.

Proof. With $Q(t)=\sum b_nn^{-it}$ we have $\sum b_n\sin(t\log n)=-\Im Q$ and $(\Im Q)^2=\frac12(|Q|^2-\Re Q^2)$. Theorem 3.5 evaluates $\int|Q|^2$, and $|\int_T^{2T}(mn)^{-it}dt|\le2/\log(mn)\le2/\log4$ bounds $\int Q^2$ by $O((\sum|b_n|)^2)$. For cos use $\Re Q$. □

Lemma 6.9. *With absolute implied constants,*

$$\int_T^{2T}D^2\,dt=\frac{T}{2\pi^2}V+O(T(1+\sqrt{V})), \tag{6.7}$$

$$\int_T^{2T}D^4\,dt\ll T(1+V)^2, \tag{6.8}$$

$$\int_T^{2T}S_X'(t)^2\,dt\ll T(1+\log X)^2. \tag{6.9}$$

Proof. All primes involved are $\le\max(X,x_0)\le T^{1/2}$. By Lemma 6.8 with $b_p=\frac{1}{\pi\sqrt{p}}$, $\int D_1^2=\frac{T}{2\pi^2}V+O(T^{1/2}+(\sum_{p\le T^{1/2}}p^{-1/2})^2)=\frac{T}{2\pi^2}V+O(T^{1/2})$. For $S_X^{(2)}(t)=-\frac{1}{2\pi}\sum_{p\le X}p^{-1}\sin(t\log p^2)$, Lemma 6.8 gives $\int(S_X^{(2)})^2\ll T\sum p^{-2}+\#\{p\le X\}+(\sum_{p\le X}p^{-1})^2\ll T$. Finally $|S_X^{(\ge3)}|\le\frac1\pi\sum_p\sum_{m\ge3}\frac{1}{mp^{m/2}}<\infty$. Expanding D^2 and using the Cauchy–Schwarz inequality for the cross terms gives (6.7).

For (6.8), $(a+b+c)^4\le27(a^4+b^4+c^4)$ reduces it to the three parts. Let $Q(t)=\sum_{p\in\mathcal{P}}p^{-1/2}p^{-it}$, so $|D_1|\le|Q|/\pi$ and $Q^2=\sum_\nu c_\nu\nu^{-it}$ with $c_{p^2}=p^{-1}$, $c_{pq}=2(pq)^{-1/2}$ for $p<q$, and $\nu\le T$. Theorem 3.5 gives $\int|Q|^4=\sum|c_\nu|^2(T+O(\nu))\ll T\sum|c_\nu|^2\le T(1+2V^2)$. For $S_X^{(2)}$ the same argument applies to $Q_2(t)=\sum_{p\le X}p^{-1}(p^2)^{-it}$, since $|S_X^{(2)}|\le|Q_2|/2\pi$: $Q_2^2=\sum d_\nu\nu^{-it}$ with $\nu=(pq)^2$, $|d_\nu|\le2(pq)^{-1}$, so $\sum|d_\nu|^2\ll1$ and $\sum\nu|d_\nu|^2\le4\#\{p\le X\}^2\le4X^2\le4T$. The part $S_X^{(\ge3)}$ is bounded.

For (6.9), $S'_X(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n)$. The terms $m = 1$ have coefficients $\frac{\log p}{\pi\sqrt{p}}$, and Lemma 6.8 with Theorem 3.6 gives $\ll T\log X \sum_{p\le X}\frac{\log p}{p} + \sum_{p\le X}\log^2 p + (\sum_{p\le X}\frac{\log p}{\sqrt{p}})^2 \ll T(1+\log X)^2 + X\log^2 X$, and $X \le T^{1/2}$. The terms $m = 2$ have coefficients $\frac{\log p}{\pi p}$ and contribute $\ll T + X\log X + \log^2 X$. The terms $m \ge 3$ are bounded by $\frac{1}{\pi}\sum_p \sum_{m\ge 3} p^{-m/2}\log p < \infty$. □

6.3. **Proof of the main theorem.** Let $E_0 = S - S^{\mathrm{pr}}_{x_0}$ and $R = S - S_X = E_0 + D$. By Theorem 3.4 and Hölder's inequality,

$$\int_T^{2T} E_0^4\,dt \ll T, \qquad \int_T^{2T} E_0^2\,dt \le T^{1/2}\Big(\int_T^{2T} E_0^4\,dt\Big)^{1/2} \ll T. \tag{6.10}$$

Write $J = \int_T^{2T} R^2\,dt$. From (6.7), (6.10) and the Cauchy–Schwarz inequality,

$$J = \frac{T}{2\pi^2}V + O(T(1+\sqrt{V})), \tag{6.11}$$

and from (6.8), (6.10) and $(a+b)^4 \le 8(a^4+b^4)$,

$$\int_T^{2T} R^4\,dt \ll T(1+V)^2. \tag{6.12}$$

Let m and n be the smallest and largest indices in $\mathcal{I}_T$ and apply (4.6). By Lemma 2.4 and (2.14), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and u_m^2, u_n^2 are $O(T^{3/4})$. The intervals $[T,\gamma_m]$ and $[\gamma_n, 2T]$ have bounded length and on them $|R|^2(\mathcal{L}+|S'_X|) \ll T^{3/4}$. Since $|\omega| \ll 1/T$, replacing $(R+\omega/\pi)^2$ by R^2 costs $\ll T^{-1}\int_T^{2T}(|R|+1)(\mathcal{L}+|S'_X|)\,dt \ll (1+V)^{1/2}\log T$ by the Cauchy–Schwarz inequality, (6.11) and (6.9). Hence

$$\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \int_T^{2T} R^2\mathcal{L}\,dt + \int_T^{2T} R^2 S'_X\,dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{6.13}$$

Since $0 \le \mathcal{L}(t) - \mathcal{L}(T) \le \frac{\log 2}{2\pi}$ on $[T,2T]$, the first integral is $\mathcal{L}(T)J + O(J)$. By the Cauchy–Schwarz inequality, (6.12) and (6.9), the second is

$$\Big|\int_T^{2T} R^2 S'_X\,dt\Big| \le \Big(\int_T^{2T} R^4\Big)^{1/2}\Big(\int_T^{2T} S'^2_X\Big)^{1/2} \ll T(1+V)(1+\log X). \tag{6.14}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (6.11),

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\,c\Big), \tag{6.15}$$

because $\log X = c\log T$. By Lemma 6.7, $(1+V)c \ll c(1+\log\frac{1}{c}) \ll 1$ for $0 < c \le \frac{1}{2}$, and $V \ll \log\log T$ because $c \ge \log 2/\log T$, so $(1+V)/\log T = o(1)$. A second application of Lemma 6.7 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2}\log\frac{1}{c}$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$. This proves (6.1). □

Remark 6.10. An alternative route to an upper bound of the form $\sum_{k\in\mathcal{I}_T} u_k(T^c)^2 \ll_c \#\mathcal{I}_T$ averages a pointwise inequality for $|u_k|$ over windows $[\gamma_k, \gamma_k + h]$ with $h = 1/\log T$, and controls the overlap of the windows through the multiplicity $M_h(t) = \#\{k : \gamma_k \le t \le \gamma_k + h\} \le 1 + N(t) - N(t-h)$, whose moments are bounded by the corollary to the Main Theorem of [6]. Such an argument needs fourth moments of $S - S_X$ and of S'_X and gives only an upper bound. The identity of Theorem 4.1 makes the window multiplicities unnecessary, needs only the second moment of S'_X, and gives the asymptotic (6.1). In the parametrization $X = K^c$ with $K \asymp T\log T$ the condition $X^2 \le T$ of the Montgomery–Vaughan step requires $c < \frac{1}{2}$; in the parametrization $X = T^c$ the endpoint $c = \frac{1}{2}$ is included.

6.4. **Localization of the zeros for small cutoffs.**

Proposition 6.11. *Let $X \geq 2$ with $M_X \leq \frac{1}{2}\mathcal{L}(T)$. Then F_X is strictly increasing on $[T, 2T]$, for every $k \in \mathcal{I}_T$ with $F_X(T) \leq k - \frac{1}{2} \leq F_X(2T)$ the truncated equation $F_X(t) = k - \frac{1}{2}$ has exactly one solution $t_k^X \in [T, 2T]$, and*

$$|\gamma_k - t_k^X| \leq \frac{2\,|u_k(X)|}{\mathcal{L}(T)}. \tag{6.16}$$

Proof. By (2.14), $F_X' \geq \mathcal{L}(T) - M_X \geq \frac{1}{2}\mathcal{L}(T)$ on $[T, 2T]$, and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$; apply the mean value theorem. □

Remark 6.12. Since $M_X \asymp \sqrt{X}$, Proposition 6.11 applies only for $X \ll \log^2 T$, where $\log(1/c) \sim \log\log T$; it then turns (6.1) into a mean-square localization of γ_k around the solutions of LeClair's truncated equation, but with a growing constant. For $X = T^c$ the bound $M_X \leq \frac{1}{2}\mathcal{L}(T)$ fails, F_X need not be monotone and the truncated equation may have several solutions near γ_k; the defect $u_k(X)$ measures the quality of the equation without selecting a solution.

7. Results conditional on a Landau-type hypothesis

Hypothesis 7.1 (L)**.** For every fixed integer $n \geq 2$, as $t \to \infty$,

$$\sum_{0<\gamma\leq t} n^{i\gamma} = -\frac{t}{2\pi}\frac{\Lambda(n)}{\sqrt{n}} + o(t), \tag{7.1}$$

the zeros being counted with multiplicity.

Proposition 7.2. *The Riemann hypothesis implies Hypothesis 7.1, with error $O(\log t)$.*

Proof. On the Riemann hypothesis $n^\rho = \sqrt{n}\,n^{i\gamma}$; apply Theorem 3.8. □

Remark 7.3. Unconditionally, Theorem 3.8 holds for $\sum n^\rho$, and $|n^\rho - \sqrt{n}\,n^{i\gamma}| \leq n\log n\,|\beta - \frac{1}{2}|$. Hypothesis 7.1 therefore follows from $\sum_{0<\gamma\leq t}|\beta - \frac{1}{2}| = o(t)$, whereas the density estimates of Selberg give $O(t)$ [19, §9.24]. Theorems 4.1–6.1 do not use Hypothesis 7.1.

Lemma 7.4. *For fixed $\beta \neq 0$, $\sum_{k\in\mathcal{I}_T} e^{i\beta\tau_k} \ll_\beta \log T$. Consequently, for fixed integers $m, n \geq 2$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} \sin(\tau_k\log m)\sin(\tau_k\log n) \longrightarrow \frac{1}{2}\,\mathbf{1}_{m=n}. \tag{7.2}$$

Proof. We may take $\beta > 0$. Let $f(k) = \beta\tau_k/2\pi$ for real k. By (2.6), $f'(k) = \beta/\log(\tau_k/2\pi)$ is positive and decreasing, and lies in $[\beta/\log(T/\pi), \frac{1}{2}]$ for $k \in \mathcal{I}_T$ and T large; hence $\|f'\| \geq \beta/\log(T/\pi)$ and Theorem 3.7 gives the bound. For (7.2) write $2\sin a\sin b = \cos(a-b) - \cos(a+b)$ and use the bound with $\beta = \log(m/n)$ and $\beta = \log(mn)$, together with $\#\mathcal{I}_T \asymp T\log T$. □

Theorem 7.5. *Assume Hypothesis 7.1. For every fixed integer $n \geq 2$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} s_k\sin(\gamma_k\log n) \longrightarrow \frac{a_n}{2} = -\frac{\Lambda(n)}{2\pi\sqrt{n}\log n}, \tag{7.3}$$

and the same holds with τ_k in place of γ_k.

Proof. Let $\alpha = \log n$ and $\Phi(t) = \int_{14}^t S(u)n^{iu}du$. With $g(u) = n^{iu}/(i\alpha)$ and $S = N - N_0 - \omega/\pi$, integration by parts gives

$$\Phi(t) = [Sg]_{14}^t - \frac{1}{i\alpha}\sum_{14<\gamma\leq t} n^{i\gamma} + \int_{14}^t g\Big(\mathcal{L} + \frac{\omega'}{\pi}\Big)du = \frac{i}{\alpha}\sum_{0<\gamma\leq t} n^{i\gamma} + O(\log t), \tag{7.4}$$

because $\int_{14}^{t} g\mathcal{L}\,du = O(\log t)$ by a second integration by parts and $\omega' = O(u^{-2})$. Hypothesis 7.1 gives $\Phi(t) = -i\,\frac{\Lambda(n)}{2\pi\sqrt{n}\,\alpha}\,t + o(t)$. By partial summation, using $\frac{d}{dt}(t\mathcal{L}(t)) = \mathcal{L}(t) + \frac{1}{2\pi}$,

$$\int_T^{2T} S(t) n^{it} \mathcal{L}(t)\,dt = \big[\Phi\mathcal{L}\big]_T^{2T} - \int_T^{2T} \frac{\Phi(t)}{2\pi t}\,dt = -i\,\frac{\Lambda(n)}{2\pi\sqrt{n}\,\alpha}\int_T^{2T} \mathcal{L}\,dt + o(T\log T). \tag{7.5}$$

Taking imaginary parts, $\int_T^{2T} S(t)\sin(\alpha t)\mathcal{L}(t)\,dt = \frac{a_n}{2}\int_T^{2T}\mathcal{L}\,dt + o(T\log T)$. Now apply Lemma 4.4 with $f(t) = \sin(\alpha t)$, $a = T$, $b = 2T$ (we may assume that T and $2T$ are not ordinates). The term with ω is $O(\log T)$, the boundary term is $O(\log^2 T)$ and $\frac{1}{2}\int_T^{2T} G^2|f'|\,dt \ll \int_T^{2T} S^2 dt + O(1) \ll T\log\log T = o(T\log T)$ by Theorem 3.3. Dividing by $\#\mathcal{I}_T = \int_T^{2T}\mathcal{L}\,dt + O(\log T)$ gives (7.3). Finally $|\sin(\alpha\tau_k) - \sin(\alpha\gamma_k)| \le \alpha|\delta_k|$ and, by (2.8) and Theorem 5.1, $\sum_{k\in\mathcal{I}_T}|s_k\delta_k| \ll (\log T)^{-1}\sum_{k\in\mathcal{I}_T} s_k^2 \ll \#\mathcal{I}_T\,\log\log T/\log T$. □

Corollary 7.6. *Assume Hypothesis 7.1. Let A be a finite set of prime powers and $\lambda \in \mathbb{R}$. Then*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(s_k^2 - \big(s_k - \lambda S_A(\tau_k)\big)^2\Big) \longrightarrow (2\lambda - \lambda^2)\,v_A, \tag{7.6}$$

with v_A as in (2.11)*. The limit is maximal for $\lambda = 1$, where it equals v_A.*

Proof. The left side is $2\lambda\langle s_k S_A(\tau_k)\rangle - \lambda^2\langle S_A(\tau_k)^2\rangle$, where $\langle\cdot\rangle$ is the mean over $\mathcal{I}_T$. By Theorem 7.5, $\langle s_k S_A(\tau_k)\rangle \to \sum_{n\in A} a_n^2/2 = v_A$, and by Lemma 7.4, $\langle S_A(\tau_k)^2\rangle \to v_A$. □

Remark 7.7. For a finite set of primes P and $A_{P,M} = \{p^m : p \in P, m \le M\}$ one has $v_{A_{P,M}} \uparrow c_P$ as $M \to \infty$ by (2.13). The experiments of Section 9 use the full local factors ($M = \infty$); the passage $M \to \infty$ inside the limit (7.6) is not covered by the proof, because the convergence in Theorem 7.5 is not shown to be uniform in n. Under the Riemann hypothesis the uniform version of Landau's formula [7] is the natural tool for that step; we do not carry it out here.

Remark 7.8. Corollary 7.6 shows that, among the multiples λS_A of the Euler phase, the Euler phase itself ($\lambda = 1$) is optimal for the clock, and that its gain v_A is half of what a perfect predictor of the corresponding Fourier components would gain. Since $v_{A_X} \to \infty$ as $X \to \infty$ while v_A is finite for every fixed A, a fixed set of primes recovers only a bounded part of the mean square $\frac{1}{2\pi^2}\log\log T$ of s_k, in agreement with Corollary 6.2.

8. The one-step corrected clock

The experiments evaluate the Euler phase at the clock point τ_k, which is known, rather than at the unknown γ_k. The following identity relates the two.

Proposition 8.1. *Let $r_k(X) = \mathcal{L}(\tau_k)\,\delta_k^X$ be the residual of the one-step clock in ordinal units. Then*

$$r_k(X) = u_k(X) + \big(S_X(\gamma_k) - S_X(\tau_k)\big) + \varepsilon_k, \qquad |\varepsilon_k| \le \frac{\delta_k^2}{2\pi\min(\gamma_k,\tau_k)}, \tag{8.1}$$

and $|S_X(\gamma_k) - S_X(\tau_k)| \le M_X|\delta_k|$.

Proof. By (2.16), $r_k(X) = \mathcal{L}(\tau_k)\delta_k - S_X(\tau_k)$, and by (2.8), $\mathcal{L}(\tau_k)\delta_k = s_k + (\mathcal{L}(\tau_k) - \mathcal{L}(\xi_k))\delta_k$ with $|\mathcal{L}(\tau_k) - \mathcal{L}(\xi_k)| \le |\delta_k|/(2\pi\min(\gamma_k,\tau_k))$. Insert $s_k = u_k(X) + S_X(\gamma_k)$. □

Heuristic 8.2. This is a heuristic argument. The middle term of (8.1) is approximately $S_X'(\gamma_k)\,s_k/\mathcal{L}(\gamma_k)$. Treating S_X', s_k and $S - S_X$ at the zeros as independent centred variables with their t-mean variances, one formally arrives at

$$\langle r_k(X)^2\rangle \approx \frac{1}{2\pi^2}\log\frac{1}{\theta} + \kappa + \theta^2\,\langle s_k^2\rangle, \qquad \theta = \frac{\log X}{\log(T/2\pi)}, \tag{8.2}$$

with a constant κ, because the t-mean of $S_X'^2$ is $\frac{1}{2\pi^2}\sum_{p\le X}\frac{\log^2 p}{p}\sim\frac{\log^2 X}{4\pi^2}=\theta^2\mathcal{L}(T)^2$. Minimizing (8.2) in θ suggests

$$\theta_{\mathrm{opt}}\approx\frac{1}{2\pi\sqrt{\langle s_k^2\rangle}},\tag{8.3}$$

which tends to 0 like $(2\log\log T)^{-1/2}$. According to this heuristic the mean square of the one-step residual with $X=T^c$ and fixed c grows like $c^2(2\pi^2)^{-1}\log\log T$, in contrast with Theorem 6.1 for u_k.

Remark 8.3. The following quantitative use of Heuristic 8.2 appears not to work. Inserting the measured $\langle s_k^2\rangle$ of Table 3 into (8.3) predicts $\theta_{\mathrm{opt}}=0.614$, 0.545, 0.508, 0.488, 0.479, 0.473 in the six windows, that is $X_{\mathrm{opt}}\approx 20$, 42, 90, 158, 214, 270. The observed optima (Table 2) are 7, 43, 151, 401, 701, 907, with least-squares exponent 0.5744. The prediction agrees in the second window and underestimates the optimal cutoff by a factor of about 3 in the top windows. The heuristic is retained because it explains why the optimal exponent of the one-step clock is a slowly varying function of the height rather than a constant, and because it separates the evaluation error $S_X(\gamma_k)-S_X(\tau_k)$, absent from Theorem 6.1, from the arithmetic defect $u_k(X)$.

Heuristic 8.4. This is a heuristic argument. In Theorem 7.5 the term $-\frac12\int G^2f'$ of (4.7) and the replacement of γ_k by τ_k are of relative size $1/\log T$. Modelling S by $\sum a_m\sin(t\log m)$, the coefficient of $\cos(t\log n)$ in S^2 for $n=p^aq^b$ with distinct primes p,q is $-a_{p^a}a_{q^b}$, and the two corrections combine to the prediction

$$\rho_n:=-2\pi\sqrt{n}\left\langle s_k\sin(\tau_k\log n)\right\rangle\approx\frac{\Lambda(n)}{\log n}+\frac{\log n}{ab\,\log(T/2\pi)}\,\mathbf{1}_{n=p^aq^b},\tag{8.4}$$

the first term being the limit given by Theorem 7.5.

9. Numerical experiments

9.1. **Data and conventions.** We use the ordinates γ_k, $1\le k\le K=2\,001\,052$, stored in double precision, and the clock values (1.1). The residuals reported are in the t-variable: δ_k for the bare clock and $\delta_k^{X,B}$ for the one-step clock with saturation level B; for fixed cutoffs X is the largest prime used. The windows of ordinals are $(0,10^3]$, $(10^3,10^4]$, $(10^4,10^5]$, $(10^5,5\cdot10^5]$, $(5\cdot10^5,10^6]$ and $(10^6,K]$. The ordinal displacement used in Tables 3 and 8 is $\mathcal{L}(\tau_k)\delta_k$, which differs from s_k by $O(\delta_k^2/\gamma_k)$. The numerics concern the one-step clock of Section 8; the defect $u_k(X)$ of Theorem 6.1 was not computed (Question 11.5).

Experiment 9.1. Object: agreement of the data with independent evaluations. Range: $k\in\{1,10^3,10^5,10^6,K,258317,258318,258319\}$ for the zeros and seven indices for the clock. Expected outcome: agreement to the precision of the data. Observed outcome: the differences between the stored ordinates and `mpmath.zetazero` are at most $3.5\cdot10^{-10}$, and the stored clock agrees with (1.1) to $3.6\cdot10^{-15}$.

9.2. **Comparison of methods.**

Experiment 9.2. Object: root mean square (RMS), maximum of |residual|, and number of residuals with $|\text{residual}|\ge\frac12$, for the bare clock, two fixed prime sets with saturation $B=0.6$, a fixed set of 303 primes without saturation, and the ordinal-dependent cutoff $X_k=\max(7,(\tau_k/2\pi)^{0.57})$ with $B=0.6$. Range: $1\le k\le K$, by windows. Expected outcome: by Proposition 5.5 fixed sets cannot change the limit superior of $|\delta_k|$ as $k\to\infty$, and by Heuristic 8.2 a fixed large cutoff is harmful at low heights. Observed outcome: Table 1 and Figures 1, 2 and 3. Over all K zeros the RMS is 0.188066 for the bare clock (maximum 0.993711 at $k=34$, 11589 residuals $\ge\frac12$), 0.120691 for $P=\{2,3,5,7\}$ (maximum 0.522347 at $k=258318$, one residual $\ge\frac12$), 0.085635 for $P=\{p\le61\}$ (maximum

0.457259 at $k = 55259$, none $\geq \frac{1}{2}$), 0.070757 for $P = \{p \leq 1999\}$ (maximum 3.247397 at $k = 1$, 156 residuals $\geq \frac{1}{2}$) and 0.066608 for the ordinal-dependent cutoff (maximum 0.513886 at $k = 1275558$, two residuals $\geq \frac{1}{2}$). In the first window the fixed set of 303 primes is worse than the bare clock (RMS 0.384555 against 0.358739), while in the top window it is close to the ordinal-dependent rule (RMS 0.064897 against 0.064259).

method and window	RMS	max |residual|	$\#\{\|\text{residual}\| \geq \frac{1}{2}\}$
bare clock τ_k			
$(0, 10^3]$	1e3]	0.358739	0.993711
$(10^3, 10^4]$	1e4]	0.274177	0.833881
$(10^4, 10^5]$	1e5]	0.226352	0.785835
$(10^5, 5 \cdot 10^5]$	5e5]	0.199294	0.762816
$(5 \cdot 10^5, 10^6]$	1e6]	0.186722	0.726355
$(10^6, 2001052]$	2001052]	0.178900	0.717359
$P = \{2, 3, 5, 7\}$, $B = 0.6$			
$(0, 10^3]$	1e3]	0.131717	0.442191
$(10^3, 10^4]$	1e4]	0.138038	0.470967
$(10^4, 10^5]$	1e5]	0.133227	0.477838
$(10^5, 5 \cdot 10^5]$	5e5]	0.125335	0.522347
$(5 \cdot 10^5, 10^6]$	1e6]	0.120648	0.484035
$(10^6, 2001052]$	2001052]	0.117413	0.489293
$P = \{p \leq 61\}$, $B = 0.6$			
$(0, 10^3]$	1e3]	0.151905	0.396903
$(10^3, 10^4]$	1e4]	0.090453	0.362670
$(10^4, 10^5]$	1e5]	0.083131	0.457259
$(10^5, 5 \cdot 10^5]$	5e5]	0.086106	0.456272
$(5 \cdot 10^5, 10^6]$	1e6]	0.085990	0.432954
$(10^6, 2001052]$	2001052]	0.085354	0.400144
$P = \{p \leq 1999\}$, *no saturation*			
$(0, 10^3]$	1e3]	0.384555	3.247397
$(10^3, 10^4]$	1e4]	0.163764	0.491263
$(10^4, 10^5]$	1e5]	0.097502	0.618127
$(10^5, 5 \cdot 10^5]$	5e5]	0.074828	0.591452
$(5 \cdot 10^5, 10^6]$	1e6]	0.068088	0.552165
$(10^6, 2001052]$	2001052]	0.064897	0.528267
$X_k = \max(7, (\tau_k/2\pi)^{0.57})$, $B = 0.6$			
$(0, 10^3]$	1e3]	0.117396	0.393711
$(10^3, 10^4]$	1e4]	0.087632	0.424347
$(10^4, 10^5]$	1e5]	0.076221	0.484853
$(10^5, 5 \cdot 10^5]$	5e5]	0.069639	0.513816
$(5 \cdot 10^5, 10^6]$	1e6]	0.066315	0.493768
$(10^6, 2001052]$	2001052]	0.064259	0.513886

TABLE 1. Residual of the bare clock and of the one-step corrected clocks, by windows of ordinals (Experiment 9.2).

9.3. **The optimal cutoff grows with the height.**

Experiment 9.3. Object: RMS of δ_k^X (no saturation) as a function of the cutoff X, for 40 cutoffs between 2 and 1999, and the minimizing cutoff in each window. Range: the six windows. Expected

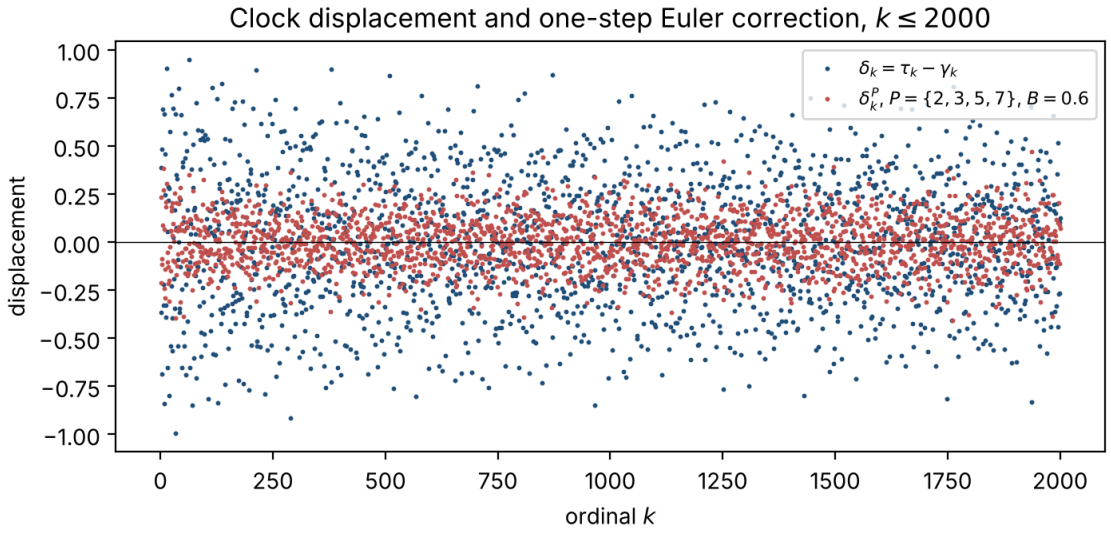


FIGURE 1. The displacement $\delta_k = \tau_k - \gamma_k$ and the residual of the one-step clock with $P = \{2, 3, 5, 7\}$ and $B = 0.6$, for $k \le 2000$.

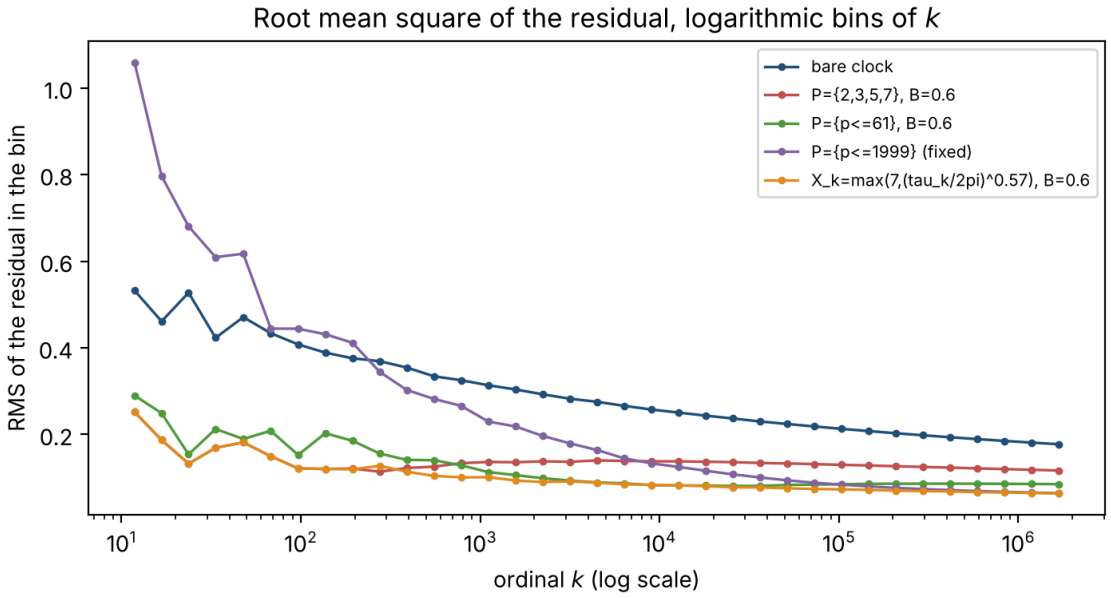


FIGURE 2. RMS of the residual in logarithmic bins of k for the five methods of Table 1.

outcome: by Corollary 6.4 the cutoff should grow with the height; Heuristic 8.2 predicts an exponent θ slowly decreasing with the height. Observed outcome: Table 2, Figures 4 and 5, and the full scan in Tables 5–6. The optimal cutoff increases from 7 to 907; the least-squares fit $\log X_{\mathrm{opt}} = \theta \log(\tau/2\pi)$ over windows 2–6 gives $\theta = 0.5744$. Near the optimum the RMS curves are flat: in the top window the RMS is 0.064467 at $X = 701$, 0.064271 at $X = 907$ and 0.064283 at $X = 1009$. We observe that the exponent is not explained quantitatively by Heuristic 8.2 (Remark 8.3).

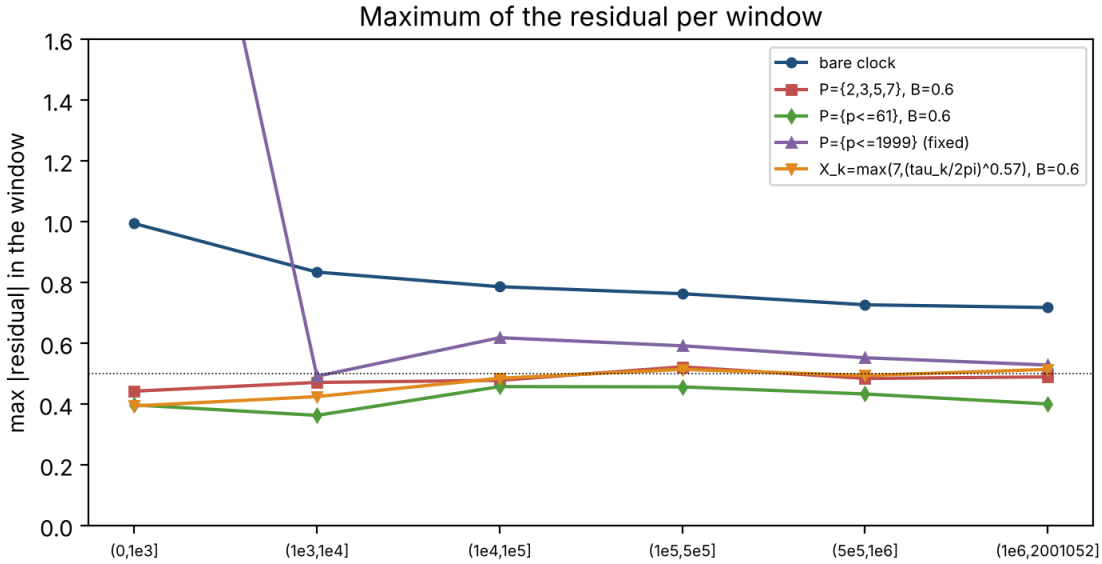


FIGURE 3. Maximum of |residual| in each window for the five methods of Table 1; the dotted line is $\frac{1}{2}$.

window	X_{opt}	RMS at X_{opt}	median $\log(\tau_k/2\pi)$	RMS of δ_k
$(0, 10^3]$	1e3]	7	0.163507	4.8617
$(10^3, 10^4]$	1e4]	43	0.089165	6.8465
$(10^4, 10^5]$	1e5]	151	0.076935	8.8540
$(10^5, 5 \cdot 10^5]$	5e5]	401	0.069779	10.3736
$(5 \cdot 10^5, 10^6]$	1e6]	701	0.066360	11.2050
$(10^6, 2001052]$	2001052]	907	0.064271	11.8382

TABLE 2. RMS-optimal prime cutoff of the one-step clock (no saturation) in each window (Experiment 9.3).

Experiment 9.4. Object: global maximum of $|\delta_k^{X,B}|$, its location, the number of residuals $\geq \frac{1}{2}$, the RMS, and the residual at $k = 258318$, for the fixed cutoffs $11 \leq X \leq 71$ and $B \in \{0.5, 0.6, 0.7\}$. Range: $1 \leq k \leq K$. Expected outcome: none from the theory, which says nothing about finite maxima (Proposition 5.5 concerns the limit). Observed outcome: Table 7. For $B = 0.5$ and $X \geq 13$ the maximum 0.493711 at $k = 34$ comes from the saturation itself: the bare residual there is 0.993711. With $B = 0.6$, the cutoffs $X \in \{13, 17, 31\}$ and all $37 \leq X \leq 71$ give a maximum below $\frac{1}{2}$ over the 2 001 052 zeros, the smallest being 0.457259 for $X = 61$.

9.4. **Mean-square gain and projections.**

Experiment 9.5. Object: the gain $\langle s_k^2 \rangle - \langle (s_k - S_P(\tau_k))^2 \rangle$ for $P = \{2, 3, 5, 7\}$ and $P = \{p \leq 61\}$ with full local factors, compared with c_P of (2.13). Range: the six windows. Expected outcome: by Corollary 7.6 and Remark 7.7 (conditional on Hypothesis 7.1), the gain of each finite truncation tends to its v_A, and $v_A \uparrow c_P$. Observed outcome: Table 3 and Figure 6. The gain increases with the height towards c_P, from 0.056957 to 0.064441 for $c_P = 0.066252$, and from 0.047822 to 0.087434 for $c_P = 0.093853$. The mean square $\langle s_k^2 \rangle$ increases from 0.067170 to 0.113224, as Theorem 5.1 predicts qualitatively; the theorem determines only the leading term $(2\pi^2)^{-1} \log\log T$ and not the constant.

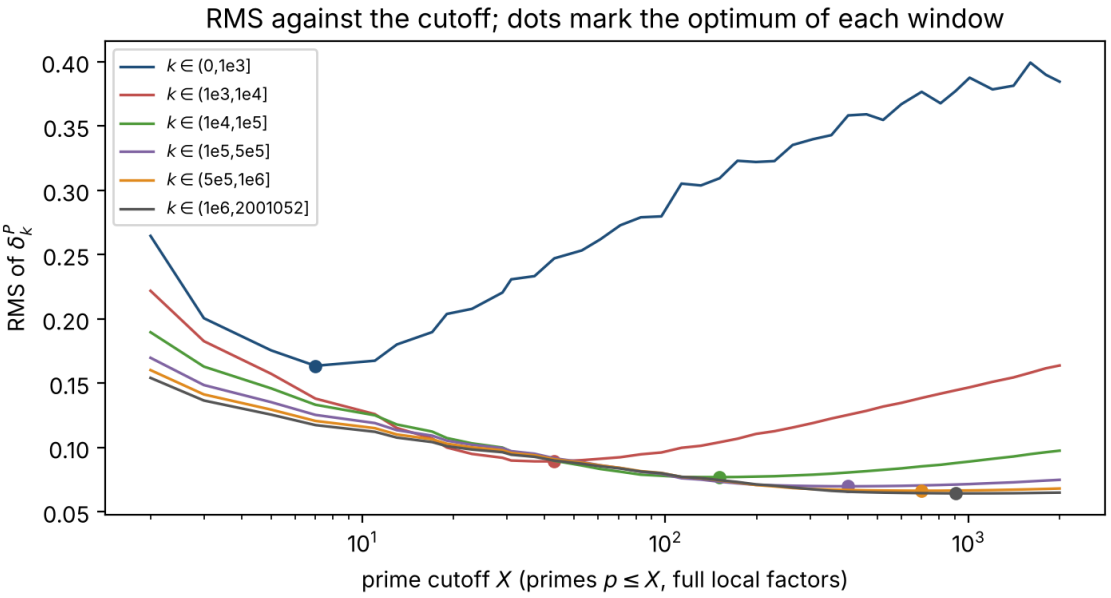


FIGURE 4. RMS of the one-step residual against the cutoff X (primes $p \le X$, full local factors), by window; dots mark the minimum of each window.

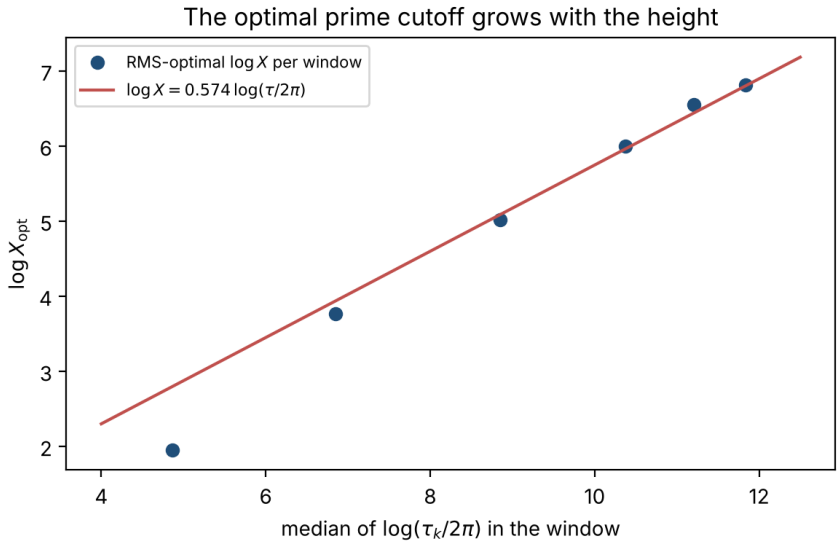


FIGURE 5. Logarithm of the optimal cutoff against the median of $\log(\tau_k/2\pi)$ in each window, with the least-squares line through the origin fitted on windows 2–6.

Experiment 9.6. Object: the normalized projections $\rho_n = -2\pi\sqrt{n}\,\langle s_k \sin(\tau_k \log n)\rangle$ for $2 \le n \le 64$. Range: $10^6 < k \le K$, where the median of $\log(\tau_k/2\pi)$ is 11.8382. Expected outcome: $\rho_n \to \Lambda(n)/\log n$ by Theorem 7.5 (conditional on Hypothesis 7.1), with the second-order correction (8.4) at finite height (Heuristic 8.4). Observed outcome: Table 8 and Figure 7. For primes ρ_p lies between 0.864 and 0.979; for prime squares between 0.526 and 0.576; over $n = p^a q^b \le 64$ the largest deviation from (8.4) is 0.0632; the largest $|\rho_n|$ over integers which are not prime powers is 0.2936, and the largest deviation from $\Lambda(n)/\log n$ over prime powers is 0.1356.

window	$\langle s_k^2 \rangle$	gain, P_7	c_{P_7}	gain, P_{61}	$c_{P_{61}}$
$(0, 10^3]$	1e3]	0.067170	0.056957	0.066252	0.047822
$(10^3, 10^4]$	1e4]	0.085195	0.063224	0.066252	0.075868
$(10^4, 10^5]$	1e5]	0.098097	0.063878	0.066252	0.084636
$(10^5, 5 \cdot 10^5]$	5e5]	0.106325	0.064185	0.066252	0.086367
$(5 \cdot 10^5, 10^6]$	1e6]	0.110476	0.064339	0.066252	0.087027
$(10^6, 2001052]$	2001052]	0.113224	0.064441	0.066252	0.087434

TABLE 3. Mean-square gain of a fixed Euler phase against its limit c_P; $P_7 = \{2, 3, 5, 7\}$, $P_{61} = \{p \le 61\}$ (Experiment 9.5).

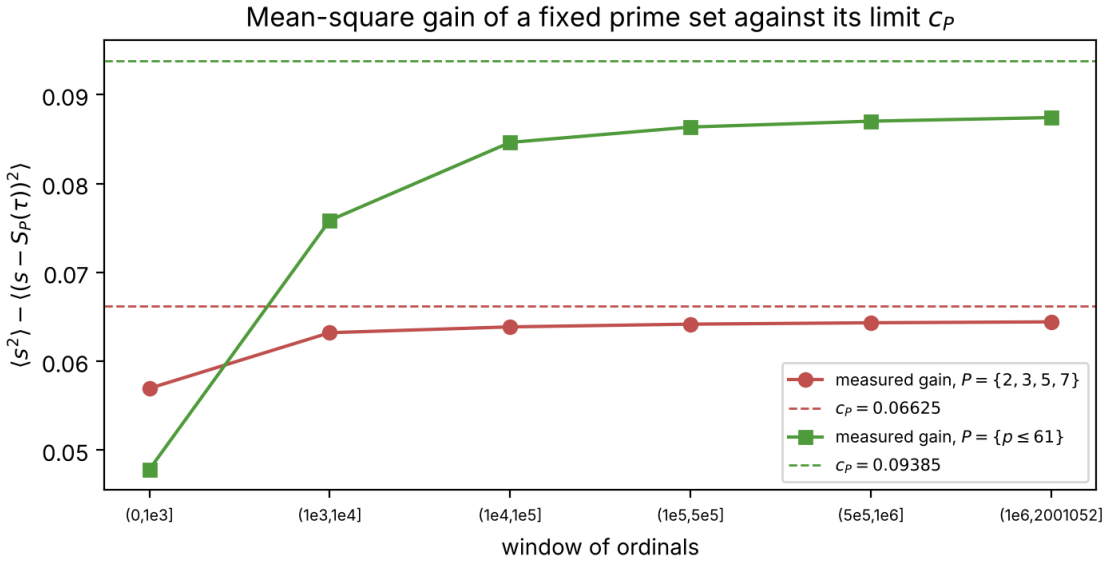


FIGURE 6. Measured mean-square gain of a fixed Euler phase in each window (markers) and the constants c_P (dashed).

9.5. **The extreme residual: a close pair of zeros.**

Experiment 9.7. Object: the zero at which the fixed set $P = \{2, 3, 5, 7\}$ with $B = 0.6$ attains its largest residual. Range: $258315 \le k \le 258321$. Expected outcome: a large $|s_k|$, which by (5.7) forces a large value of $|S|$ nearby. Observed outcome: Table 4 and Figure 8. The ordinates $\gamma_{258317} = 175681.649131916$ and $\gamma_{258318} = 175681.715894456$ form a close pair with normalized gap 0.1088 (rank 2371 among 2 001 051 gaps), followed by a normalized gap 2.6587. The midpoint value is $S(\gamma_{258318}) = 1.0394$ and $\delta_{258318} = 0.6379$. The residuals at $k = 258318$ are 0.5223 ($P = \{2, 3, 5, 7\}$, $B = 0.6$), 0.2345 ($P = \{p \le 61\}$, $B = 0.6$), 0.3373 ($P = \{p \le 1999\}$) and 0.2031 (ordinal-dependent cutoff).

10. DISCUSSION

10.1. **Status of the results.** Theorems 4.1, 5.1 and 6.1, their corollaries, and Propositions 5.4, 5.5, 6.11 and 8.1 are unconditional. Theorem 7.5 and Corollary 7.6 assume Hypothesis 7.1, which follows from the Riemann hypothesis. Heuristics 8.2 and 8.4 are heuristic. The value 0.57 of the exponent and all statements about finite maxima are numerical observations.

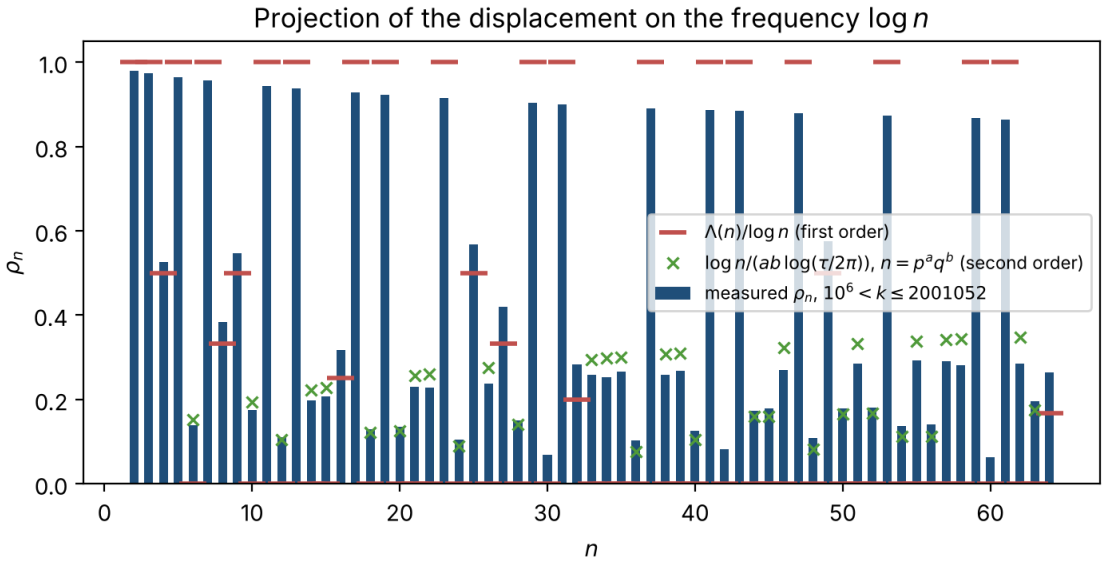


FIGURE 7. Normalized projections ρ_n of the displacement on $\sin(\tau_k \log n)$, $10^6 < k \le K$, with the first-order limit $\Lambda(n)/\log n$ and the second-order prediction (8.4).

k	γ_k	τ_k	δ_k	$\gamma_{k+1}-\gamma_k$	normalized gap
258315	175680.155201189	175680.512721900	+0.3575	0.8010	1.3053
258316	175680.956210354	175681.126401291	+0.1702	0.6929	1.1291
258317	175681.649131916	175681.740080472	+0.0909	0.0668	0.1088
258318	175681.715894456	175682.353759444	+0.6379	1.6316	2.6587
258319	175683.347455659	175682.967438207	−0.3800	0.5181	0.8443
258320	175683.865589247	175683.581116760	−0.2845	0.4856	0.7914
258321	175684.351235152	175684.194795104	−0.1564	0.3002	0.4892

TABLE 4. Ordinates, clock and gaps around the close pair $k = 258317, 258318$; the normalized gap is $(\gamma_{k+1}-\gamma_k)\log(\gamma_k/2\pi)/2\pi$ (Experiment 9.7).

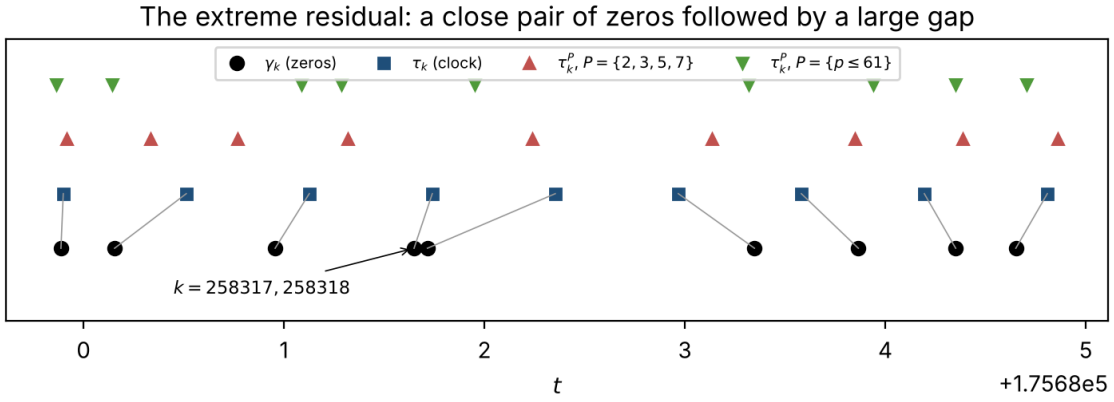


FIGURE 8. Zeros, clock and one-step clocks around the close pair $k = 258317, 258318$.

10.2. **What is proved about the improvement.** Theorem 6.1 answers the question whether a number of primes depending on the ordinal improves the truncated França–LeClair equation, in the mean-square sense and for the equation itself: a fixed number of primes leaves a defect whose mean square grows like $(2\pi^2)^{-1}\log\log T$, a cutoff $X = T^c$ keeps it bounded, and boundedness requires $\log X \asymp \log T$. The power law is therefore forced by the theorem; the value of the exponent is not. Within $0 < c \le \frac{1}{2}$ the bound (6.1) decreases with c, which suggests taking c as large as the method allows for the equation itself. For the one-step clock, which evaluates the phase at τ_k, the optimal exponent results from a balance described by Heuristic 8.2, and no theorem is proved.

The sup norm behaves differently. In the t-variable, $\limsup|\delta_k| \le 0.6331$ unconditionally and $\delta_k \to 0$ on the Riemann hypothesis (Proposition 5.4); a fixed set of primes changes no limit superior (Proposition 5.5). The improvement of finite maxima seen in Experiment 9.4 is therefore a phenomenon at finite height.

10.3. **Novelty.** Selberg's approximation of $S(t)$ by prime sums up to a power of T is a statement on t-averages and is due to Selberg; Selberg and Korolev treat the integer displacement from Gram intervals (Remark 5.3). We have not found in the literature we consulted the identity of Theorem 4.1, the mean-square theorem for the defect at the zeros (Theorem 6.1) or the dichotomy of Corollary 6.4; the absence of results in a search is weak evidence, and these statements may have antecedents that we missed. In [13] the number of primes is subject to $N(t) \to \infty$ and $N < [t^2]$, without a rule selecting it from the ordinal, and the truncated equation is supported by a probabilistic model; we found no proof of its validity in [4, 5, 13]. For finite Euler products near the critical line in a different direction see Gonek [8].

Remark 10.1. The results proved here concern the arithmetic part of the displacement, of size $\sqrt{\log\log T}$, and its removal by primes; they do not address the local spacing statistics of the zeros. For instance, replacing each γ_k by $\gamma_k + \epsilon_k$ with $|\epsilon_k| \le \epsilon/\mathcal{L}(\gamma_k)$ changes each s_k by at most $\epsilon(1+o(1))$, hence the root mean square in Theorem 5.1 by at most $\epsilon(1+o(1))$, whatever the local statistics of the perturbed sequence. In particular Theorem 5.1 does not imply random matrix statistics for the zeros.

11. Open questions and future work

Question 11.1. Does $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(T^c)^2 = \frac{1}{2\pi^2}\log\frac{1}{c} + \kappa + o(1)$ hold for fixed $c \in (0, \frac{1}{2}]$, with an explicit constant κ?

Question 11.2. Does Theorem 6.1 extend to $\frac{1}{2} < c \le 2$, which contains LeClair's range $N < [t^2]$? The proof uses $X^2 \le T$ in the fourth-moment step of Lemma 6.9.

Question 11.3. Does (6.1) hold for cutoffs $X_k = (\tau_k/2\pi)^\theta$ that vary from zero to zero? This needs a maximal inequality for partial sums of S_X at the zeros.

Question 11.4. Is the mean square of the one-step residual $r_k(T^c)$ asymptotic to $c^2(2\pi^2)^{-1}\log\log T$ for fixed c, as Heuristic 8.2 suggests, and what is the optimal exponent $\theta(T)$ of the one-step clock?

Question 11.5. Future experiment: compute $u_k(X)$ for $X = 2^{rc}$ on the windows $\mathcal{I}_{2^r}$ and compare with (6.1); compute the solutions of $F_X(t) = k - \frac{1}{2}$ by Newton's method for $X \ll \log^2 T$ and compare with Proposition 6.11.

Question 11.6. Possible generalization: Dirichlet L-functions, for which França and LeClair give the analogous equation [5] and Selberg's approximation is available in the form of [6, §3] for primitive characters.

Appendix A. Complete tables

Table 5: RMS of δ_k^X (no saturation) for all cutoffs of the scan of Experiment 9.3.

X	$(0, 10^3]$	$(10^3, 10^4]$	$(10^4, 10^5]$	$(10^5, 5 \cdot 10^5]$	$(5 \cdot 10^5, 10^6]$	$(10^6, K]$
2	0.264585	0.221853	0.189628	0.169743	0.160181	0.154125
3	0.200546	0.182692	0.162903	0.148543	0.141254	0.136519
5	0.175536	0.157325	0.145975	0.135250	0.129435	0.125551
7	0.163507	0.138018	0.133227	0.125335	0.120648	0.117413
11	0.167491	0.125933	0.125019	0.118978	0.115014	0.112199
13	0.180112	0.115143	0.117861	0.113445	0.110121	0.107676
17	0.189732	0.107018	0.112357	0.109187	0.106362	0.104196
19	0.203848	0.099967	0.107341	0.105291	0.102932	0.101027
23	0.207822	0.094970	0.103192	0.102058	0.100080	0.098398
29	0.220519	0.091980	0.099933	0.099516	0.097835	0.096324
31	0.230788	0.089845	0.096818	0.097113	0.095707	0.094360
37	0.233278	0.089201	0.094209	0.095126	0.093936	0.092724
43	0.247108	0.089165	0.089653	0.091595	0.090813	0.089816
53	0.253280	0.090105	0.085984	0.088604	0.088177	0.087369
61	0.261896	0.091200	0.083253	0.086131	0.085995	0.085356
71	0.272840	0.092404	0.081157	0.083975	0.084099	0.083607
83	0.279043	0.094679	0.078846	0.081114	0.081573	0.081281
97	0.279760	0.096097	0.077984	0.079519	0.080174	0.079986
113	0.305237	0.099735	0.077086	0.076058	0.077117	0.077155
131	0.303814	0.101313	0.076968	0.075024	0.076116	0.076235
151	0.309424	0.104066	0.076935	0.073336	0.074335	0.074579
173	0.323001	0.106809	0.077044	0.072094	0.072794	0.073149
199	0.322057	0.110538	0.077336	0.070915	0.070796	0.071304
229	0.322766	0.112658	0.077638	0.070557	0.069704	0.070261
263	0.335293	0.115531	0.078195	0.070231	0.068448	0.068864
307	0.339800	0.119016	0.078840	0.069985	0.067633	0.067458
353	0.342957	0.122519	0.079635	0.069831	0.067253	0.066262
401	0.358302	0.125367	0.080542	0.069779	0.066961	0.065561
461	0.359151	0.128498	0.081570	0.069855	0.066669	0.065176
523	0.354759	0.131831	0.082601	0.069960	0.066526	0.064877
601	0.367055	0.134709	0.083735	0.070147	0.066431	0.064663
701	0.376638	0.138525	0.085322	0.070484	0.066360	0.064467
809	0.367728	0.141873	0.086526	0.070817	0.066390	0.064353
907	0.377272	0.144454	0.087902	0.071167	0.066521	0.064271
1009	0.387648	0.146798	0.089073	0.071525	0.066595	0.064283
1201	0.378526	0.151049	0.091182	0.072272	0.066845	0.064317
1409	0.381402	0.154503	0.092992	0.072896	0.067167	0.064412
1601	0.399367	0.158134	0.094832	0.073618	0.067469	0.064584
1801	0.389913	0.161587	0.096305	0.074283	0.067767	0.064768
1999	0.384555	0.163764	0.097502	0.074828	0.068088	0.064897

Table 6: Maximum of $|\delta_k^X|$ (no saturation) for all cutoffs of the scan of Experiment 9.3.

X	$(0, 10^3]$	$(10^3, 10^4]$	$(10^4, 10^5]$	$(10^5, 5 \cdot 10^5]$	$(5 \cdot 10^5, 10^6]$	$(10^6, K]$
2	0.714732	0.769491	0.692832	0.720658	0.661936	0.671779
3	0.778082	0.590169	0.597440	0.601584	0.561789	0.568719

X	$(0, 10^3]$	$(10^3, 10^4]$	$(10^4, 10^5]$	$(10^5, 5 \cdot 10^5]$	$(5 \cdot 10^5, 10^6]$	$(10^6, K]$
5	1.374655	0.516964	0.542554	0.529087	0.499111	0.519284
7	1.363459	0.470967	0.477838	0.522347	0.484035	0.489293
11	1.508302	0.509751	0.476272	0.484013	0.461281	0.472383
13	1.890769	0.471351	0.485959	0.459791	0.471622	0.445972
17	2.030160	0.462473	0.484965	0.471862	0.452001	0.438833
19	2.579012	0.511457	0.473939	0.436459	0.420326	0.419404
23	2.086207	0.480575	0.516961	0.449248	0.426305	0.396139
29	2.532059	0.517976	0.486233	0.456782	0.425924	0.421632
31	2.731378	0.478395	0.503207	0.436347	0.394374	0.408531
37	2.436264	0.472277	0.467938	0.457662	0.402816	0.408334
43	2.920237	0.445443	0.482769	0.454351	0.425856	0.418435
53	2.844272	0.421161	0.474280	0.499673	0.435596	0.405677
61	2.717402	0.386775	0.457259	0.456272	0.432954	0.400144
71	3.238857	0.440040	0.447583	0.474087	0.436888	0.399374
83	2.963768	0.484372	0.467632	0.463983	0.438356	0.404333
97	2.894415	0.492464	0.483027	0.471593	0.460266	0.426854
113	3.849430	0.505889	0.513140	0.459718	0.483898	0.444630
131	3.439317	0.529451	0.517859	0.460881	0.484771	0.422286
151	3.367629	0.513247	0.525446	0.470577	0.482027	0.452456
173	3.996564	0.501890	0.549235	0.495148	0.456277	0.461665
199	3.365175	0.506558	0.523619	0.495310	0.458465	0.468141
229	3.335133	0.493618	0.528107	0.497048	0.468220	0.469958
263	4.072437	0.509804	0.557233	0.535397	0.479418	0.457026
307	3.875649	0.526799	0.546887	0.526425	0.467465	0.473187
353	3.465466	0.541694	0.561048	0.534833	0.473423	0.477070
401	4.309564	0.517594	0.555640	0.563591	0.504123	0.520443
461	4.028754	0.524278	0.544453	0.564897	0.521484	0.515423
523	3.255435	0.547997	0.577784	0.573315	0.492713	0.513253
601	4.111242	0.510670	0.585122	0.569968	0.507595	0.499887
701	4.359529	0.561843	0.576930	0.548586	0.495677	0.503530
809	3.399704	0.551479	0.599065	0.553709	0.486320	0.525568
907	4.022965	0.520528	0.588646	0.540603	0.524873	0.529711
1009	4.759390	0.506597	0.574853	0.553235	0.523729	0.521027
1201	3.414545	0.491973	0.602652	0.560114	0.551342	0.534354
1409	3.934751	0.472026	0.587566	0.572365	0.566273	0.533096
1601	4.812450	0.524598	0.593689	0.572861	0.554297	0.526446
1801	3.551103	0.528815	0.586851	0.576451	0.554530	0.526215
1999	3.247397	0.491263	0.618127	0.591452	0.552165	0.528267

Table 7: Fixed cutoffs and saturation levels over all K zeros (Experiment 9.4).

X	B	global max	k at max	$\# \geq \frac{1}{2}$	RMS	residual at $k = 258318$
11	0.5	0.497093	1252	0	0.114954	+0.468830
11	0.6	0.509751	1252	1	0.114954	+0.468830
11	0.7	0.509751	1252	1	0.114954	+0.468830
13	0.5	0.493711	34	0	0.109968	+0.414562
13	0.6	0.485959	28074	0	0.109974	+0.414562
13	0.7	0.485959	28074	0	0.109975	+0.414562
17	0.5	0.493711	34	0	0.106136	+0.377680

X	B	global max	k at max	$\# \geq \frac{1}{2}$	RMS	residual at $k = 258318$
17	0.6	0.484965	28075	0	0.106150	+0.377680
17	0.7	0.484965	28075	0	0.106153	+0.377680
19	0.5	0.493711	34	0	0.102639	+0.351715
19	0.6	0.511457	4988	1	0.102665	+0.351715
19	0.7	0.511457	4988	1	0.102670	+0.351715
23	0.5	0.493711	34	0	0.099739	+0.321005
23	0.6	0.516961	28075	1	0.099777	+0.321005
23	0.7	0.516961	28075	1	0.099784	+0.321005
29	0.5	0.493711	34	0	0.097456	+0.320746
29	0.6	0.517976	4988	1	0.097509	+0.320746
29	0.7	0.517976	4988	1	0.097518	+0.320746
31	0.5	0.493711	34	0	0.095297	+0.286603
31	0.6	0.481293	28075	0	0.095364	+0.286603
31	0.7	0.503207	31812	1	0.095375	+0.286603
37	0.5	0.493711	34	0	0.093503	+0.257718
37	0.6	0.472277	6709	0	0.093584	+0.257718
37	0.7	0.481833	7	0	0.093598	+0.257718
41	0.5	0.493711	34	0	0.091884	+0.282516
41	0.6	0.480998	5444	0	0.091977	+0.282516
41	0.7	0.481833	7	0	0.091994	+0.282516
43	0.5	0.493711	34	0	0.090330	+0.255696
43	0.6	0.482769	55259	0	0.090438	+0.255696
43	0.7	0.482769	55259	0	0.090456	+0.255696
47	0.5	0.493711	34	0	0.088909	+0.233942
47	0.6	0.477323	30265	0	0.089031	+0.233942
47	0.7	0.489035	31812	0	0.089052	+0.233942
53	0.5	0.493711	34	0	0.087660	+0.260451
53	0.6	0.474829	100674	0	0.087797	+0.260451
53	0.7	0.499673	100674	0	0.087821	+0.260451
59	0.5	0.493711	34	0	0.086551	+0.239402
59	0.6	0.474829	100674	0	0.086700	+0.239402
59	0.7	0.482408	100674	0	0.086726	+0.239402
61	0.5	0.493711	34	0	0.085471	+0.234496
61	0.6	0.457259	55259	0	0.085635	+0.234496
61	0.7	0.481833	7	0	0.085663	+0.234496
67	0.5	0.493711	34	0	0.084496	+0.258256
67	0.6	0.474829	100674	0	0.084673	+0.258256
67	0.7	0.481833	7	0	0.084704	+0.258256
71	0.5	0.493711	34	0	0.083579	+0.248477
71	0.6	0.458509	100674	0	0.083769	+0.248477
71	0.7	0.481833	7	0	0.083803	+0.248477

Table 8: Projections of the displacement, $10^6 < k \leq K$ (Experiment 9.6). Columns: n; $\langle \mathcal{L}(\tau_k)\delta_k \sin(\tau_k \log n)\rangle$; ρ_n; $\Lambda(n)/\log n$; second-order prediction $\log n/(ab\log(\tau/2\pi))$ for $n = p^a q^b$ with the median of $\log(\tau_k/2\pi)$.

n	projection	ρ_n	$\Lambda(n)/\log n$	second order
2	$-1.101798 \cdot 10^{-1}$	0.979032	1.000000	–
3	$-8.948284 \cdot 10^{-2}$	0.973824	1.000000	–

n	projection	ρ_n	$\Lambda(n)/\log n$	second order
4	$-4.184672 \cdot 10^{-2}$	0.525861	0.500000	–
5	$-6.870881 \cdot 10^{-2}$	0.965333	1.000000	–
6	$-9.026866 \cdot 10^{-3}$	0.138929	0.000000	0.151353
7	$-5.760368 \cdot 10^{-2}$	0.957589	1.000000	–
8	$-2.155326 \cdot 10^{-2}$	0.383035	0.333333	–
9	$-2.905212 \cdot 10^{-2}$	0.547620	0.500000	–
10	$-8.834723 \cdot 10^{-3}$	0.175539	0.000000	0.194504
11	$-4.529915 \cdot 10^{-2}$	0.943987	1.000000	–
12	$-5.057500 \cdot 10^{-3}$	0.110079	0.000000	0.104952
13	$-4.140890 \cdot 10^{-2}$	0.938092	1.000000	–
14	$-8.442033 \cdot 10^{-3}$	0.198468	0.000000	0.222926
15	$-8.499428 \cdot 10^{-3}$	0.206831	0.000000	0.228754
16	$-1.262904 \cdot 10^{-2}$	0.317402	0.250000	–
17	$-3.580933 \cdot 10^{-2}$	0.927685	1.000000	–
18	$-4.855880 \cdot 10^{-3}$	0.129445	0.000000	0.122078
19	$-3.369338 \cdot 10^{-2}$	0.922787	1.000000	–
20	$-4.806896 \cdot 10^{-3}$	0.135070	0.000000	0.126528
21	$-7.977501 \cdot 10^{-3}$	0.229698	0.000000	0.257177
22	$-7.712575 \cdot 10^{-3}$	0.227295	0.000000	0.261106
23	$-3.034556 \cdot 10^{-2}$	0.914406	1.000000	–
24	$-3.431327 \cdot 10^{-3}$	0.105620	0.000000	0.089486
25	$-1.807767 \cdot 10^{-2}$	0.567927	0.500000	–
26	$-7.409792 \cdot 10^{-3}$	0.237396	0.000000	0.275218
27	$-1.284469 \cdot 10^{-2}$	0.419359	0.333333	–
28	$-4.547679 \cdot 10^{-3}$	0.151199	0.000000	0.140739
29	$-2.670938 \cdot 10^{-2}$	0.903738	1.000000	–
30	$-1.996010 \cdot 10^{-3}$	0.068692	0.000000	–
31	$-2.574513 \cdot 10^{-2}$	0.900649	1.000000	–
32	$-7.950979 \cdot 10^{-3}$	0.282602	0.200000	–
33	$-7.157940 \cdot 10^{-3}$	0.258360	0.000000	0.295357
34	$-6.894815 \cdot 10^{-3}$	0.252605	0.000000	0.297879
35	$-7.143158 \cdot 10^{-3}$	0.265524	0.000000	0.300327
36	$-2.718760 \cdot 10^{-3}$	0.102495	0.000000	0.075677
37	$-2.332464 \cdot 10^{-2}$	0.891447	1.000000	–
38	$-6.689950 \cdot 10^{-3}$	0.259116	0.000000	0.307274
39	$-6.836467 \cdot 10^{-3}$	0.268253	0.000000	0.309468
40	$-3.176432 \cdot 10^{-3}$	0.126226	0.000000	0.103869
41	$-2.204080 \cdot 10^{-2}$	0.886746	1.000000	–
42	$-2.004407 \cdot 10^{-3}$	0.081619	0.000000	–
43	$-2.145466 \cdot 10^{-2}$	0.883966	1.000000	–
44	$-4.148809 \cdot 10^{-3}$	0.172914	0.000000	0.159829
45	$-4.254161 \cdot 10^{-3}$	0.179308	0.000000	0.160778
46	$-6.334462 \cdot 10^{-3}$	0.269941	0.000000	0.323413
47	$-2.040786 \cdot 10^{-2}$	0.879076	1.000000	–
48	$-2.483491 \cdot 10^{-3}$	0.108109	0.000000	0.081752
49	$-1.309455 \cdot 10^{-2}$	0.575928	0.500000	–
50	$-4.016163 \cdot 10^{-3}$	0.178433	0.000000	0.165228
51	$-6.336008 \cdot 10^{-3}$	0.284303	0.000000	0.332129
52	$-3.977664 \cdot 10^{-3}$	0.180223	0.000000	0.166885
53	$-1.907514 \cdot 10^{-2}$	0.872540	1.000000	–
54	$-2.978449 \cdot 10^{-3}$	0.137520	0.000000	0.112319
55	$-6.301149 \cdot 10^{-3}$	0.293617	0.000000	0.338507

n	projection	ρ_n	$\Lambda(n)/\log n$	second order
56	$-3.008650\cdot10^{-3}$	0.141464	0.000000	0.113343
57	$-6.113419\cdot10^{-3}$	0.290002	0.000000	0.341525
58	$-5.863989\cdot10^{-3}$	0.280600	0.000000	0.342994
59	$-1.796165\cdot10^{-2}$	0.866866	1.000000	–
60	$-1.282342\cdot10^{-3}$	0.062411	0.000000	–
61	$-1.761495\cdot10^{-2}$	0.864423	1.000000	–
62	$-5.768611\cdot10^{-3}$	0.285395	0.000000	0.348627
63	$-3.940823\cdot10^{-3}$	0.196534	0.000000	0.174989
64	$-5.250276\cdot10^{-3}$	0.263908	0.166667	–

Appendix B. Reproducibility

The repository accompanying this paper contains `code/opc_analysis.py` (Python 3 with NumPy, SciPy, Matplotlib and mpmath), which produces every table and figure of Section 9 in about one minute and is reproduced in full in Appendix C after the bibliography, the outputs in `outputs/`, and the figures in `figures/`. The data files `zeross.bin` (ordinates) and `zLamb.bin` (clock), each 2 001 052 little-endian double-precision numbers, are not included in the repository because of their size; they are expected in `data/`.

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Appendix C. Python code

The following script, `code/opc_analysis.py`, produces all tables and figures of Section 9 from the data files described in Appendix B.

```
"""Ordinal-dependent prime cutoffs for the Franca--LeClair approximation of the Riemann zeros.

Numerical companion of the manuscript ordinal_prime_cutoff_v1.

Data (not included in the repository, see README):
  data/zeross.bin : ordinates gamma_k of the zeros of zeta, k = 1..2,001,052 (little-endian float64)
  data/zLamb.bin  : Lambert clock tau_k = 2*pi*(k-11/8)/W((k-11/8)/e), same length (little-endian float64)

Notation (as in the manuscript):
  delta_k   = tau_k - gamma_k                    (displacement of the clock)
  T'(k)     = 2*pi / log(tau_k/(2*pi))           (derivative of the clock in the ordinal)
  S_P(t)    = -(1/pi) sum_{p in P} arg(1 - p^{-1/2-it})     (Euler phase, full local factors)
  tau_k^P   = tau_k - T'(k) S_P(tau_k)           (one-step prime-corrected clock)
  delta_k^P = tau_k^P - gamma_k = delta_k - T'(k) S_P(tau_k)
  saturation with level B: the correction T'(k) S_P(tau_k) is replaced by sgn(.) min(|.|, B).

Outputs: ../outputs/*.csv, *.txt, *.tex and ../figures/*.png
Usage:   python3 opc_analysis.py [path_to_data_dir]
"""
import os
import sys
import time

import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import mpmath as mp
from scipy.special import lambertw

T0 = time.time()
HERE = os.path.dirname(os.path.abspath(__file__))
DATA = sys.argv[1] if len(sys.argv) > 1 else os.path.join(HERE, "..", "data")
OUT = os.path.join(HERE, "..", "outputs")
FIG = os.path.join(HERE, "..", "figures")
os.makedirs(OUT, exist_ok=True)
os.makedirs(FIG, exist_ok=True)
LOG = open(os.path.join(OUT, "opc_analysis_log.txt"), "w")


def log(*a):
    s = " ".join(str(x) for x in a)
    print(s)
    LOG.write(s + "\n")
    LOG.flush()
```

```
# ------------------------------------------------------------------------------ data
tau = np.fromfile(os.path.join(DATA, "zLamb.bin"), dtype="<f8")
gam = np.fromfile(os.path.join(DATA, "zeross.bin"), dtype="<f8")
K = min(len(tau), len(gam))
tau, gam = tau[:K], gam[:K]
k = np.arange(1, K + 1, dtype=float)
Lk = np.log(tau / (2 * np.pi))          # log(tau_k / 2pi) = 1 + W_k
Tp = 2 * np.pi / Lk                     # T'(k)
delta = tau - gam                       # delta_k = tau_k - gamma_k
s_ord = delta / Tp                      # ordinal-normalized displacement
log("K =", K, " tau_K =", tau[-1], " gamma_K =", gam[-1])

# check of the clock formula on a sample of indices
idx = np.array([0, 1, 9, 99, 999, 99999, K - 1])
a = idx + 1 - 11.0 / 8.0
Wv = np.real(lambertw(a / np.e))
tau_chk = 2 * np.pi * a / Wv
log("clock formula check, max |tau_file - 2pi(k-11/8)/W((k-11/8)/e)| on sample:",
    float(np.max(np.abs(tau_chk - tau[idx]))))

EDGES = [0, 1000, 10**4, 10**5, 5 * 10**5, 10**6, K]
WIN = list(zip(EDGES[:-1], EDGES[1:]))
WLAB = ["(0,1e3]", "(1e3,1e4]", "(1e4,1e5]", "(1e5,5e5]", "(5e5,1e6]", "(1e6,%d]" % K]


def rms(x):
    return float(np.sqrt(np.mean(x * x)))


def primes_upto(n):
    sv = np.ones(n + 1, bool)
    sv[:2] = False
    for q in range(2, int(n ** 0.5) + 1):
        if sv[q]:
            sv[q * q::q] = False
    return np.nonzero(sv)[0]


def euler_term(p, t):
    """-(1/pi) arg(1 - p^{-1/2-it}): contribution of the prime p to S_P(t), all powers p^m"""
    return -np.angle(1.0 - p ** -0.5 * np.exp(-1j * t * np.log(p))) / np.pi


def saturate(c, B):
    return np.sign(c) * np.minimum(np.abs(c), B) if B is not None else c


def li2(x, terms=200):
    m = np.arange(1, terms + 1)
    return float(np.sum(x ** m / m ** 2))


def cP(plist):
    """(1/2pi^2) sum_{p in P} Li_2(1/p): asymptotic mean-square gain of the full Euler phase over P"""
    return sum(li2(1.0 / p) for p in plist) / (2 * np.pi ** 2)


PR = primes_upto(2000)
LADDER = [p for p in PR if p in (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 43, 53, 61, 71, 83, 97, 113, 131,
                                 151, 173, 199, 229, 263, 307, 353, 401, 461, 523, 601, 701, 809, 907, 1009,
    1201,
                                 1409, 1601, 1801, 1999)]
SCAN = [p for p in PR if 11 <= p <= 71]
```

```
THETA = 0.57                                        # variable rule X_k = max(7, (tau_k/2pi)^THETA)
Xvar = np.maximum(7.0, (tau / (2 * np.pi)) ** THETA)
B0 = 0.6

# ------------------------------------------------------------------------------ one pass over the primes
S = np.zeros(K)          # S_P(tau_k) with P = primes <= current p
Svar = np.zeros(K)       # variable cutoff: prime p used for k with X_k >= p
snap = {}
rms_ladder = {}          # p -> list of RMS per window (no saturation)
max_ladder = {}
scan_rows = []
for p in PR:
    tp = euler_term(p, tau)
    S += tp
    a0 = int(np.searchsorted(Xvar >= p, True)) if Xvar[-1] >= p else K
    if a0 < K:
        Svar[a0:] += tp[a0:]
    if p in LADDER:
        r = delta - Tp * S
        rms_ladder[p] = [rms(r[a:b]) for a, b in WIN]
        max_ladder[p] = [float(np.max(np.abs(r[a:b]))) for a, b in WIN]
    if p in SCAN:
        for B in (0.5, 0.6, 0.7):
            r = delta - saturate(Tp * S, B)
            ar = np.abs(r)
            j = int(np.argmax(ar))
            scan_rows.append((p, B, float(ar[j]), j + 1, int(np.sum(ar >= 0.5)), rms(r), float(r[258317])))
    if p in (7, 61, 1999):
        snap[p] = S.copy()
log("prime loop done, %.0f s" % (time.time() - T0))

# ------------------------------------------------------------------------------ Table A: methods by window
methods = [("bare clock", np.zeros(K), None),
           ("P={2,3,5,7}, B=0.6", Tp * snap[7], B0),
           ("P={p<=61}, B=0.6", Tp * snap[61], B0),
           ("P={p<=1999} (fixed)", Tp * snap[1999], None),
           ("X_k=max(7,(tau_k/2pi)^0.57), B=0.6", Tp * Svar, B0)]
resid = {}
with open(os.path.join(OUT, "table_A_methods_by_window.csv"), "w") as f:
    f.write("method,window,RMS,max_abs,count_ge_half\n")
    for name, c, B in methods:
        r = delta - saturate(c, B)
        resid[name] = r
        for (a, b), wl in zip(WIN, WLAB):
            f.write('"%s",%s,%.6f,%.6f,%d\n' % (name, wl, rms(r[a:b]), np.max(np.abs(r[a:b])),
                                                np.sum(np.abs(r[a:b]) >= 0.5)))
        j = int(np.argmax(np.abs(r)))
        log("A| %-38s global RMS %.6f  max %.6f at k=%d  #>=1/2: %d" %
            (name, rms(r), abs(r[j]), j + 1, int(np.sum(np.abs(r) >= 0.5))))
        log("   per-window RMS:", " ".join("%.4f" % rms(r[a:b]) for a, b in WIN))
        log("   per-window max:", " ".join("%.4f" % np.max(np.abs(r[a:b])) for a, b in WIN))

# ------------------------------------------------------------------------------ Table B: RMS against the
    cutoff
with open(os.path.join(OUT, "table_B_rms_vs_cutoff.csv"), "w") as f:
    f.write("X," + ",".join("RMS_" + w for w in WLAB) + "," + ",".join("max_" + w for w in WLAB) + "\n")
    for p in LADDER:
        f.write("%d," % p + ",".join("%.6f" % v for v in rms_ladder[p]) + "," +
                ",".join("%.6f" % v for v in max_ladder[p]) + "\n")
opt = []
for j, ((a, b), wl) in enumerate(zip(WIN, WLAB)):
    pbest = min(LADDER, key=lambda p: rms_ladder[p][j])
    Lmed = float(np.median(Lk[a:b]))
    opt.append((wl, pbest, rms_ladder[pbest][j], Lmed, rms(delta[a:b])))
    log("B| window %-16s optimal X = %5d  RMS = %.4f   median log(tau/2pi) = %.3f  bare RMS = %.4f" %
```

```
        (wl, pbest, rms_ladder[pbest][j], Lmed, rms(delta[a:b])))
xs = np.array([o[3] for o in opt[1:]])
ys = np.log(np.array([o[1] for o in opt[1:]], float))
theta_fit = float(np.sum(xs * ys) / np.sum(xs * xs))
log("B| least-squares fit log X_opt = theta * log(tau/2pi) on windows 2-6: theta = %.4f" % theta_fit)
with open(os.path.join(OUT, "table_B_optimal_cutoff.csv"), "w") as f:
    f.write("window,X_opt,RMS_at_opt,median_log_tau_over_2pi,bare_RMS\n")
    for o in opt:
        f.write("%s,%d,%.6f,%.4f,%.6f\n" % o)
    f.write("theta_fit,%.6f\n" % theta_fit)

# ---------------------------------------------------------------------------- Table C: fixed cutoff and
    saturation, global
with open(os.path.join(OUT, "table_C_fixed_cutoff_saturation.csv"), "w") as f:
    f.write("X,B,global_max,k_at_max,count_ge_half,RMS,residual_at_k258318\n")
    for row in scan_rows:
        f.write("%d,%.1f,%.6f,%d,%d,%.6f,%.6f\n" % row)
for row in scan_rows:
    if row[0] in (11, 13, 31, 61, 71):
        log("C| X=%3d B=%.1f  max %.4f at k=%8d  #>=1/2 %d  RMS %.4f  r(258318) %+.4f" % row)

# ---------------------------------------------------------------------------- Table D: measured mean-square
    gain against theory
PP = {7: [2, 3, 5, 7], 61: [int(p) for p in PR if p <= 61], 1999: [int(p) for p in PR if p <= 1999]}
with open(os.path.join(OUT, "table_D_gain_vs_theory.csv"), "w") as f:
    f.write("window,mean_s2,gain_P7,c_P7,gain_P61,c_P61\n")
    for (a, b), wl in zip(WIN, WLAB):
        s2 = float(np.mean(s_ord[a:b] ** 2))
        g7 = s2 - float(np.mean((s_ord[a:b] - snap[7][a:b]) ** 2))
        g61 = s2 - float(np.mean((s_ord[a:b] - snap[61][a:b]) ** 2))
        f.write("%s,%.6f,%.6f,%.6f,%.6f,%.6f\n" % (wl, s2, g7, cP(PP[7]), g61, cP(PP[61])))
        log("D| %-16s <s^2> %.4f  gain P7 %.4f (c=%.4f)  gain P61 %.4f (c=%.4f)" %
            (wl, s2, g7, cP(PP[7]), g61, cP(PP[61])))

# ---------------------------------------------------------------------------- Table E: projection
    coefficients
a, b = 10**6, K
def Lambda(n):
    for p in PR:
        if p > n:
            break
        if n % p == 0:
            m = n
            while m % p == 0:
                m //= p
            return float(np.log(p)) if m == 1 else 0.0
    return 0.0

Lmed_top = float(np.median(Lk[a:b]))


def factor(n):
    out = {}
    for p in PR:
        if p > n:
            break
        while n % p == 0:
            out[p] = out.get(p, 0) + 1
            n //= p
    return out


def second_order(n):
    """second-order prediction log(n)/(a*b*log(tau/2pi)) for n = p^a q^b with two distinct primes, else nan"""
    fz = factor(n)
```

```
    if len(fz) != 2:
        return float("nan")
    e = list(fz.values())
    return float(np.log(n)) / (e[0] * e[1] * Lmed_top)


rows_E = []
for n in range(2, 65):
    proj = float(np.mean(s_ord[a:b] * np.sin(tau[a:b] * np.log(n))))
    rho = -2 * np.pi * np.sqrt(n) * proj
    theo = Lambda(n) / np.log(n)
    rows_E.append((n, proj, rho, theo, second_order(n)))
with open(os.path.join(OUT, "table_E_projection_coefficients.csv"), "w") as f:
    f.write("n,projection,rho_n=-2pi*sqrt(n)*projection,Lambda(n)/log(n),second_order_log(n)/(ab*L)\n")
    for r_ in rows_E:
        f.write("%d,%.6e,%.6f,%.6f,%.6f\n" % r_)
so_err = [abs(r_[2] - r_[4]) for r_ in rows_E if r_[4] == r_[4]]
log("E| median log(tau/2pi) in the window: %.4f ; max |rho_n - log n/(ab L)| over n = p^a q^b <= 64: %.4f" %
    (Lmed_top, max(so_err)))
log("E| n, rho_n, first-order, second-order:", " ".join("%d:%.3f/%.3f/%.3f" % (r_[0], r_[2], r_[3], r_[4])
                                                        for r_ in rows_E))
pp_err = [abs(r_[2] - r_[3]) for r_ in rows_E if r_[3] > 0]
np_err = [abs(r_[2]) for r_ in rows_E if r_[3] == 0]
log("E| window k in (1e6, K]: max |rho_n - Lambda(n)/log n| over prime powers n<=64: %.4f ; max |rho_n| over"
    " non-prime-powers: %.4f" % (max(pp_err), max(np_err)))
log("E| rho_2..rho_12:", " ".join("%d:%.3f" % (r_[0], r_[2]) for r_ in rows_E[:11]))

# ------------------------------------------------------------------------------ the extreme residual: a close
    pair
kk = 258318
i0 = kk - 1
ng = (gam[1:] - gam[:-1]) * np.log(gam[:-1] / (2 * np.pi)) / (2 * np.pi)
rank = int(np.sum(ng < ng[i0 - 1])) + 1
mp.mp.dps = 25
with open(os.path.join(OUT, "close_pair_k258318.txt"), "w") as f:
    f.write("k, gamma_k (zeross.bin), tau_k, delta_k=tau-gamma, gap to next, normalized gap to next\n")
    for j in range(i0 - 3, i0 + 4):
        f.write("%d %.9f %.9f %+.4f %.4f %.4f\n" % (j + 1, gam[j], tau[j], delta[j], gam[j + 1] - gam[j], ng[j
    ]))
    for n in (kk - 1, kk, kk + 1):
        z = mp.zetazero(n)
        f.write("mpmath zetazero(%d) = %s ; difference with zeross.bin = %.3e\n" %
                (n, mp.nstr(z.imag, 18), float(z.imag) - gam[n - 1]))
    f.write("normalized gap (k-1,k) = %.4f, rank %d among %d gaps\n" % (ng[i0 - 1], rank, len(ng)))
    for n in (1, 1000, 100000, 1000000, K):
        z = mp.zetazero(n)
        f.write("data check: mpmath zetazero(%d) - zeross.bin = %.3e\n" % (n, float(z.imag) - gam[n - 1]))
    f.write("S(gamma_k) (midpoint) = %s\n" % mp.nstr(kk - mp.mpf(1) / 2 - mp.siegeltheta(mp.mpf(gam[i0])) / mp.
    pi - 1, 10))
    for name in resid:
        f.write("residual %-38s %+.4f\n" % (name, resid[name][i0]))
log(open(os.path.join(OUT, "close_pair_k258318.txt")).read())

# ------------------------------------------------------------------------------ figures
plt.rcParams.update({"font.size": 10, "figure.dpi": 200, "savefig.dpi": 200})
C0, C1, C2, C3, C4 = "#1f4e79", "#c0504d", "#4f9a3a", "#8064a2", "#e08a1e"

# fig01: displacement for k <= 2000
fig, ax = plt.subplots(figsize=(7.2, 3.6))
n0 = 2000
ax.plot(k[:n0], delta[:n0], ".", ms=2.2, color=C0, label=r"$\delta_k=\tau_k-\gamma_k$")
ax.plot(k[:n0], resid["P={2,3,5,7}, B=0.6"][:n0], ".", ms=2.2, color=C1,
        label=r"$\delta_k^{P}$, $P=\{2,3,5,7\}$, $B=0.6$")
ax.axhline(0, color="k", lw=0.5)
ax.set_xlabel("ordinal $k$")
```

```
ax.set_ylabel("displacement")
ax.legend(loc="upper right", fontsize=8)
ax.set_title("Clock displacement and one-step Euler correction, $k\\leq 2000$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig01_displacement_first2000.png"))

# fig02: binned RMS against log10 k
bins = np.unique(np.round(np.logspace(1, np.log10(K), 36)).astype(int))
mid, curves = [], {nm: [] for nm in resid}
for a_, b_ in zip(bins[:-1], bins[1:]):
    mid.append(np.sqrt(a_ * b_))
    for nm in resid:
        curves[nm].append(rms(resid[nm][a_:b_]))
fig, ax = plt.subplots(figsize=(7.2, 4.0))
for (nm, col) in zip(resid, (C0, C1, C2, C3, C4)):
    ax.plot(mid, curves[nm], "-o", ms=2.5, lw=1.2, color=col, label=nm)
ax.set_xscale("log")
ax.set_xlabel("ordinal $k$ (log scale)")
ax.set_ylabel("RMS of the residual in the bin")
ax.legend(fontsize=7.5)
ax.set_title("Root mean square of the residual, logarithmic bins of $k$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig02_rms_by_height.png"))

# fig03: RMS against the cutoff, per window
fig, ax = plt.subplots(figsize=(7.2, 4.0))
cols = (C0, C1, C2, C3, C4, "#555555")
for j, wl in enumerate(WLAB):
    yv = [rms_ladder[p][j] for p in LADDER]
    ax.plot(LADDER, yv, "-", lw=1.2, color=cols[j], label="$k\\in$" + wl)
    pb = opt[j][1]
    ax.plot([pb], [opt[j][2]], "o", color=cols[j], ms=5)
ax.set_xscale("log")
ax.set_xlabel("prime cutoff $X$ (primes $p\\leq X$, full local factors)")
ax.set_ylabel("RMS of $\\delta_k^{P}$")
ax.legend(fontsize=7.5)
ax.set_title("RMS against the cutoff; dots mark the optimum of each window")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig03_rms_vs_cutoff.png"))

# fig04: optimal cutoff against log(tau/2pi)
fig, ax = plt.subplots(figsize=(6.0, 4.0))
xo = np.array([o[3] for o in opt])
yo = np.log(np.array([o[1] for o in opt], float))
ax.plot(xo, yo, "o", color=C0, label="RMS-optimal $\\log X$ per window")
xx = np.linspace(4, 12.5, 50)
ax.plot(xx, theta_fit * xx, "-", color=C1, label="$\\log X=%.3f\\,\\log(\\tau/2\\pi)$" % theta_fit)
ax.set_xlabel("median of $\\log(\\tau_k/2\\pi)$ in the window")
ax.set_ylabel("$\\log X_{\\rm opt}$")
ax.legend(fontsize=8)
ax.set_title("The optimal prime cutoff grows with the height")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig04_optimal_cutoff.png"))

# fig05: projection coefficients
fig, ax = plt.subplots(figsize=(7.2, 3.8))
nn = np.array([r_[0] for r_ in rows_E])
rr = np.array([r_[2] for r_ in rows_E])
th = np.array([r_[3] for r_ in rows_E])
ax.bar(nn, rr, width=0.6, color=C0, label="measured $\\rho_n$, $10^6<k\\leq %d$" % K)
ax.plot(nn, th, "_", color=C1, ms=12, mew=2, label="$\\Lambda(n)/\\log n$ (first order)")
so = np.array([r_[4] for r_ in rows_E])
ax.plot(nn, so, "x", color=C2, ms=5, mew=1.2, label="$\\log n/(ab\\,\\log(\\tau/2\\pi))$, $n=p^aq^b$ (second
    order)")
ax.axhline(0, color="k", lw=0.5)
```

```
ax.set_xlabel("$n$")
ax.set_ylabel("$\\rho_n$")
ax.legend(fontsize=8)
ax.set_title("Projection of the displacement on the frequency $\\log n$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig05_projection_coefficients.png"))

# fig06: close pair
fig, ax = plt.subplots(figsize=(7.2, 2.8))
js = np.arange(i0 - 4, i0 + 5)
ax.plot(gam[js], np.zeros_like(js), "o", color="k", ms=6, label="$\\gamma_k$ (zeros)")
ax.plot(tau[js], 0.6 * np.ones_like(js), "s", color=C0, ms=5, label="$\\tau_k$ (clock)")
ax.plot(tau[js] - (Tp * snap[7])[js], 1.2 * np.ones_like(js), "^", color=C1, ms=5, label="$\\tau_k^P$, $P
    =\\{2,3,5,7\\}$")
ax.plot(tau[js] - (Tp * snap[61])[js], 1.8 * np.ones_like(js), "v", color=C2, ms=5, label="$\\tau_k^P$, $P=\\{
    p\\leq61\\}$")
for j in js:
    ax.plot([gam[j], tau[j]], [0, 0.6], "-", color="#999999", lw=0.6)
ax.annotate("$k=258317,258318$", xy=(gam[i0 - 1], 0), xytext=(gam[i0 - 1] - 1.2, -0.5), fontsize=8,
            arrowprops=dict(arrowstyle="->", lw=0.6))
ax.set_yticks([])
ax.set_ylim(-0.8, 2.3)
ax.set_xlabel("$t$")
ax.legend(fontsize=7, ncol=4, loc="upper center")
ax.set_title("The extreme residual: a close pair of zeros followed by a large gap")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig06_close_pair.png"))

# fig07: measured gain against theory
g7s, g61s, s2s = [], [], []
for a_, b_ in WIN:
    s2 = float(np.mean(s_ord[a_:b_] ** 2))
    s2s.append(s2)
    g7s.append(s2 - float(np.mean((s_ord[a_:b_] - snap[7][a_:b_]) ** 2)))
    g61s.append(s2 - float(np.mean((s_ord[a_:b_] - snap[61][a_:b_]) ** 2)))
fig, ax = plt.subplots(figsize=(7.2, 3.8))
xw = np.arange(len(WLAB))
ax.plot(xw, g7s, "o-", color=C1, label="measured gain, $P=\\{2,3,5,7\\}$")
ax.axhline(cP(PP[7]), color=C1, ls="--", lw=1, label="$c_P=%.5f$" % cP(PP[7]))
ax.plot(xw, g61s, "s-", color=C2, label="measured gain, $P=\\{p\\leq 61\\}$")
ax.axhline(cP(PP[61]), color=C2, ls="--", lw=1, label="$c_P=%.5f$" % cP(PP[61]))
ax.set_xticks(xw)
ax.set_xticklabels(WLAB, fontsize=7)
ax.set_xlabel("window of ordinals")
ax.set_ylabel("$\\langle s^2\\rangle-\\langle (s-S_P(\\tau))^2\\rangle$")
ax.legend(fontsize=7.5)
ax.set_title("Mean-square gain of a fixed prime set against its limit $c_P$")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig07_gain_vs_theory.png"))

# fig08: maximum per window
fig, ax = plt.subplots(figsize=(7.2, 3.8))
for (nm, col, mk) in zip(resid, (C0, C1, C2, C3, C4), "osd^v"):
    ax.plot(xw, [np.max(np.abs(resid[nm][a_:b_])) for a_, b_ in WIN], "-" + mk, color=col, ms=4, label=nm)
ax.axhline(0.5, color="k", lw=0.6, ls=":")
ax.set_xticks(xw)
ax.set_xticklabels(WLAB, fontsize=7)
ax.set_ylabel("max $|$residual$|$ in the window")
ax.set_ylim(0, 1.6)
ax.legend(fontsize=7)
ax.set_title("Maximum of the residual per window")
fig.tight_layout()
fig.savefig(os.path.join(FIG, "fig08_max_by_window.png"))

log("total time %.0f s" % (time.time() - T0))
```

```
LOG.close()
```